

BACHARELADO EM
CIÊNCIA DA COMPUTAÇÃO

**MARKET REACTIONS TO UNITED STATES-CHINA
TARIFF WAR EVENTS: AN EVENT STUDY AND DATA
SCIENCE APPROACH**

JOÃO PEDRO DUARTE MÓL

Brasília - DF, 2026

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Trabalho de Conclusão de Curso apresentado como requisito parcial para a obtenção de grau de Bacharel em Ciência da Computação, pelo Instituto Brasileiro de Ensino, Desenvolvimento e Pesquisa (IDP).

Orientadora

MSc. Lorena de Souza Bezerra Borges

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Aprovado em

Banca Examinadora

MSc. Lorena de Souza Bezerra Borges - Orientadora

Me. Lucas Mauricio Castro e Martins - Examinador Interno

Esp. Edvard Rodrigo Sousa Corrêa - Examinador Interno

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ABSTRACT

This thesis examines how the U.S. stock market, chiefly the S&P 500, responded to key episodes of the United States–China tariff war. The framing is deliberate: rather than asserting that tariffs *caused* particular financial outcomes, the study asks whether tariff-related information shocks coincided in time with abnormal market behavior, and whether any such pattern survives changes in the event window, the sample, and the estimation method. Talking in terms of *temporal association* is a methodological choice, not a hedge: isolating a clean causal effect would require ruling out the many confounding macroeconomic forces that move equity prices around the same dates, which a single-country event study cannot credibly do.

The empirical design pairs a data-science–oriented Event Study with volatility analysis, robustness and sensitivity checks, a placebo test, and an exploratory Machine Learning layer. A master chronology of 67 tariff-war events was assembled for documentary purposes, but the financial inference draws only on episodes with usable market data: a core set of 17 events from the classical phase of the dispute and an expanded set of 24 market-covered events; episodes from 2021 onward fall outside the period with continuous data and are excluded from the inferential results. From daily series for the S&P 500, Nasdaq Composite, Shanghai Composite, and VIX, the analysis constructed returns, event windows, abnormal returns, cumulative abnormal returns (CAR), and volatility measures.

The evidence points to associations that are weak, uneven, and context-dependent. Mean CAR turned mildly negative over the shorter windows and slightly positive over the longest one, yet no test reached significance at the 5% level, and every bootstrap confidence interval straddled zero. A robust-estimation layer (comprising the median, trimmed and winsorized means, a sign test, and leave-one-out diagnostics) showed that the mean-based reading hinges on individual high-impact observations, most visibly event E08 in the $[-3, +3]$ window, while the placebo test left the observed CAR values hard to distinguish from randomly chosen non-event windows. The exploratory Machine Learning models failed to beat a simple majority baseline, underscoring how little of the market's reaction is directly predictable from so small a set of events. The contribution is therefore less a verdict than a careful accounting: this thesis documents the limits and heterogeneity of market-based evidence on tariff-war shocks and supplies a reproducible empirical pipeline linking historical policy events to financial behavior, deliberately stopping short of claiming more than the data will bear.

Keywords: U.S.-China Tariff War, Event Study, Cumulative Abnormal Returns, Trade Policy Uncertainty, Financial Markets, Data Science, Machine Learning.

RESUMO

Esta monografia examina como o mercado acionário norte-americano, sobretudo o S&P 500, reagiu a episódios-chave da guerra tarifária entre Estados Unidos e China. O enquadramento é deliberado: em vez de afirmar que as tarifas *causaram* resultados financeiros, o estudo investiga se choques informacionais ligados às tarifas coincidiram no tempo com comportamento anormal do mercado, e se algum padrão dessa natureza resiste a variações na janela de evento, na amostra e no método de estimação. Falar em *associação temporal* é uma escolha metodológica, não uma evasiva: isolar um efeito causal limpo exigiria descartar os muitos fatores macroeconômicos concorrentes que movem os preços nas mesmas datas, algo que um estudo de evento centrado em um único país não consegue fazer de forma crível.

O desenho empírico combina um Estudo de Evento de orientação *data science* com análise de volatilidade, testes de robustez e sensibilidade, um teste placebo e uma camada exploratória de *Machine Learning*. Uma cronologia mestra de 67 eventos foi reunida com finalidade documental, mas a inferência recorre apenas aos episódios com dados de mercado utilizáveis: um conjunto central de 17 eventos da fase clássica da disputa e um ampliado de 24 eventos com cobertura de mercado; os episódios a partir de 2021 ficam fora do período com dados contínuos e são excluídos da inferência. A partir das séries diárias do S&P 500, Nasdaq Composite, Shanghai Composite e VIX, foram construídos retornos, janelas de evento, retornos anormais, retornos anormais acumulados (CAR) e medidas de volatilidade.

A evidência aponta para associações fracas, irregulares e dependentes de contexto. O CAR médio foi levemente negativo nas janelas curtas e ligeiramente positivo na mais longa, mas nenhum teste alcançou significância a 5% e todos os intervalos de confiança por *bootstrap* abrangeram o zero. Uma camada de estimação robusta (composta pela mediana, médias aparada e winsorizada, teste de sinais e diagnósticos *leave-one-out*) mostrou que a leitura pela média depende de observações individuais de alto impacto, com destaque para o evento E08 na janela $[-3, +3]$, enquanto o teste placebo tornou os valores de CAR difíceis de distinguir de janelas sem evento escolhidas ao acaso. Os modelos de *Machine Learning* não superaram uma linha de base de maioria simples, reforçando o quão pouco da reação do mercado é diretamente previsível a partir de tão poucos eventos. A contribuição é, portanto, menos um veredito do que uma prestação de contas cuidadosa: documentam-se os limites e a heterogeneidade da evidência de mercado sobre choques tarifários, oferecendo uma estrutura empírica reproduzível que conecta política a comportamento financeiro sem afirmar mais do que os dados sustentam.

Palavras-chave: Guerra Tarifária EUA-China, Estudo de Eventos, Retornos Anormais Acumulados, Incerteza de Política Comercial, Mercados Financeiros, Ciência de Dados, Aprendizado de Máquina.

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INTRODUCTION

The escalation of the United States–China trade dispute since 2018 introduced substantial uncertainty into global financial markets and raised important questions about how trade policy shocks are incorporated into asset prices. As the two largest economies in the world, the United States and China occupy central positions in global production networks, international supply chains, investor expectations, and capital flows. Consequently, tariff policy changes between these economies have implications that extend beyond bilateral trade volumes, affecting corporate margins, risk perception, and financial market behavior. The 2018 tariff conflict affected U.S. prices and welfare [1], while broader evidence on economic [2] and trade policy uncertainty [3] shows that uncertainty itself can operate as a relevant macro-financial shock.

Trade policy uncertainty is particularly relevant because financial markets respond not only to implemented tariff measures, but also to expectations regarding future restrictions, retaliation, negotiation progress, and possible policy reversals. In this sense, tariff announcements, implementations, suspensions, reductions, and trade agreements may be interpreted as observable information shocks. These events can influence investor expectations regarding future trade costs, supply chain disruptions, firm profitability, and geopolitical risk. Prior research on the U.S.–China trade war suggests that trade linkages and firm exposure can affect firm value [4], while tariff-related uncertainty can influence market valuation [5] and investor behavior. Related evidence at the industry and company level also suggests that U.S. industries and firms were unevenly exposed to trade uncertainty [6] and tariff-related costs [7]. Collectively, this body of evidence motivates a systematic empirical framework for evaluating market reactions around specific policy events.

The central financial indicator in this research is the Standard & Poor's 500 index. As the primary benchmark for large-cap U.S. equities, the S&P 500 tracks over 500 leading firms representing approximately 80% of total available market capitalization [8], with roughly 13.5 trillion U.S. dollars invested in vehicles that track or benchmark against it [8]. Because the index spans firms with diverse sensitivities to global trade, technology, and consumer behavior, it captures broad equity market sentiment rather than conditions in any single sector, making it suitable for detecting aggregate reactions to tariff-war events. The Nasdaq Composite, the Shanghai Composite, and the VIX com-

plement the primary indicator by offering perspectives on technology-oriented firms, Chinese market conditions, and implied volatility, respectively.

The empirical scope of the research is deliberately delimited. The project documents a broader master chronology of tariff-war events, but financial inference is restricted to events with sufficient market coverage and valid event-window construction. This boundary matters because preserving historical completeness and overstating the evidence available for market-based inference are different goals, and conflating them would obscure what the data can actually support. The thesis therefore separates documented historical events from the subset that can be empirically analyzed through daily financial data.

The empirical method is the Event Study framework. Event studies are widely used in empirical finance to examine how asset prices react to new information [9] within pre-defined event windows [10–12]. The U.S.–China tariff war unfolded through discrete announcements, implementation dates, suspensions, reductions, and negotiation milestones that map naturally onto daily financial market data, making it a suitable candidate for this approach. By estimating expected returns, calculating abnormal returns, and aggregating them into Cumulative Abnormal Returns (CAR), the framework permits a structured assessment of whether market behavior around tariff-war events deviated from the pre-event baseline.

Beyond the Event Study framework, the project adopts data science practices to improve transparency, verifiability, and methodological consistency. Implemented in Python, the pipeline constructs a structured chronology of tariff-war events, separates them into historical regimes, engineers financial features from daily market data, generates event windows, calculates abnormal returns and CAR, and runs robustness and sensitivity analyses. All procedures are documented so the workflow can be repeated, inspected, and validated, following principles of computational reproducibility [13], scientific computing [14], and incremental testing [15]. The complete source code, datasets, and figures are publicly available at <https://github.com/joaopdmol/TCC---U.S.-CHINA-TARIFF-WAR-EVENTS>.

An exploratory Machine Learning layer complements the Event Study. Logistic Regression and Random Forest are used not as the primary evidentiary basis, but to test whether financial features derived from event windows contain enough information to classify the directional sign of market reaction. Random Forest draws on Breiman [16] and broader statistical learning references [17, 18]. Given the small number of event-window observations and the inherent noise in financial returns data, this layer is interpreted with restraint throughout [19].

1.1 RESEARCH PROBLEM AND QUESTION

Although tariffs and trade policy uncertainty have been widely examined in the eco-

economic literature, it remains challenging to assess how financial markets respond to individual tariff-war events. The United States–China tariff war did not occur as a single isolated shock. Instead, it developed through a sequence of announcements, implementations, suspensions, negotiations, and retaliatory measures. Because of this structure, market reactions may depend not only on the content of each event, but also on investor expectations, prior pricing, timing, and the broader macroeconomic environment in which the event occurred.

This thesis does not claim to establish strict econometric causality. Instead, it evaluates temporal associations between historically dated tariff-war events and financial market reactions. The research question guiding this study is:

How did the U.S. stock market react to key events of the United States–China tariff war, and to what extent are these market reactions, measured through Cumulative Abnormal Returns across specified event windows and macroeconomic regimes, stable under robustness and sensitivity testing?

1.2 OBJECTIVES

The general objective of this research is to examine market reactions to key events within the United States–China tariff war in the U.S. stock market through an Event Study framework supported by data science practices, robustness analysis, and an exploratory Machine Learning layer. The objective is not to estimate definitive causal financial impacts, but to evaluate whether tariff-war events were temporally associated with abnormal market behavior across different event windows and sample specifications.

The specific objectives are:

1. To construct a historical chronology of major tariff-war events, classified according to escalation and relief signals.
2. To quantify abnormal returns and Cumulative Abnormal Returns surrounding these events using predefined event windows.
3. To document the separation of events into core, pandemic, and post-pandemic regimes, distinguishing the main core sample from the expanded market-covered sample.
4. To evaluate the robustness of market reactions through sensitivity analysis, alternative estimation windows, signal thresholds, and sample composition checks.
5. To explore whether financial features derived from event windows can classify the directional sign of market reaction using Machine Learning as an exploratory complement to the Event Study.

1.3 CONTRIBUTIONS AND STRUCTURE

This thesis contributes to empirical finance and applied data science by organizing the analysis of tariff-war events into a structured and reproducible research pipeline. The contribution is not based on model complexity alone, but on the integration of a classical Event Study design with documented event selection, market data processing, robustness checks, volatility analysis, and a limited exploratory Machine Learning component. This design allows the study to evaluate market reactions while preserving a clear distinction between measurable association, statistical uncertainty, and causal interpretation.

Importantly, this thesis also contributes by explicitly documenting instances of mature-market inertia: periods in which widely tracked indices fail to exhibit statistically significant or systematic reactions around well-defined information shocks. The deliberate reporting of these null findings is treated as a contribution in its own right, since it counterbalances the well-known publication bias favoring statistically positive results in empirical finance [20] and provides a transparent empirical baseline against which future analyses of comparable trade-policy episodes can be benchmarked.

The remainder of this thesis is organized as follows. Chapter 2 reviews the literature on Trade Policy Uncertainty, financial market reactions to information shocks, Event Study methodology, data science reproducibility, and Machine Learning in financial analysis. Chapter 3 presents the methodological framework, including event chronology construction, market data integration, event window design, expected return estimation, abnormal returns, CAR, robustness analysis, and exploratory Machine Learning. Chapter 4 reports the empirical results. Chapter 5 discusses the economic and methodological implications of the findings. Chapter 6 concludes the thesis by summarizing the main contributions, limitations, and future research directions.

2

2

LITERATURE REVIEW

2.1 THEORETICAL FRAMEWORK

Four interconnected bodies of literature underpin the empirical design of this research. The first covers the core concepts of trade policy uncertainty, the U.S.–China tariff war, and how financial markets respond to information shocks. The second addresses Event Study methodology as the empirical framework for measuring abnormal returns and cumulative abnormal returns around discrete policy events. The third introduces the Machine Learning models used as an exploratory complement. The fourth reviews related empirical work on the financial effects of the tariff conflict. Together, these bodies of knowledge contribute distinct layers to the analytical design: trade policy uncertainty theory motivates treating tariff-war events as information shocks; market efficiency literature supports using asset prices as proxies of investor response; the Event Study provides the formal measurement framework; and Machine Learning contributes a supplementary diagnostic of directional predictability.

2.1.1 Conceptual Definitions and Scope

Several key concepts recur throughout the thesis and are worth defining at the outset. A tariff is a tax or duty imposed on imported goods, usually designed to protect domestic industries, alter trade incentives, or respond to perceived unfair trade practices. A trade war occurs when countries impose successive restrictive trade measures against each other, often through tariffs, retaliatory tariffs, quotas, or other barriers. In the context of this research, the United States–China tariff war is treated as a sequence of historically dated policy events rather than as a single isolated shock.

Tariff-war events are classified here into two broad analytical groups. Escalation events refer to announcements, implementations, increases, retaliatory actions, or other measures that intensify trade tensions. Relief events refer to suspensions, reductions, agreements, negotiation progress, or other signals that temporarily reduce trade tensions. These categories are analytical, not moral: they provide a repeatable structure for comparing market responses to different types of trade-policy signals.

Trade Policy Uncertainty (TPU) refers to uncertainty about the direction, intensity, timing, or permanence of trade-related policy decisions. Such uncertainty matters in fi-

financial markets because investors price not only current economic conditions, but also expectations about future costs, profitability, supply chain disruptions, and macroeconomic risk. Tariff announcements and negotiation signals can therefore operate as information shocks even before their full economic effects materialize.

A financial market reaction, for the purposes of this research, refers to observable changes in market behavior around tariff-war events, measured through returns, abnormal returns, cumulative abnormal returns, and volatility. The S&P 500 serves as the main market indicator because it provides a broad measure of U.S. equity market performance. Accordingly, the analysis is not an estimate of the tariff war's full welfare effect, but an examination of how aggregate equity prices moved around specific tariff-related information events.

2.1.2 Trade Policy Uncertainty and the U.S.–China Tariff War

The relationship between geopolitical developments, trade policy uncertainty, and financial market reactions has become a central topic in empirical finance. Economic Policy Uncertainty provides a useful theoretical foundation for this discussion because it captures how ambiguity regarding future policy decisions can affect investment, expectations, and asset prices. The work of Baker, Bloom, and Davis established a widely used framework for measuring policy uncertainty [2] and demonstrated that uncertainty can operate as a relevant macroeconomic and financial shock. In the context of international trade, Caldara et al. show that trade policy uncertainty can generate measurable economic effects, especially when policy announcements and retaliatory measures alter expectations about future trade conditions [3].

The U.S.–China tariff war represents a particularly relevant case for studying this mechanism. Beginning in 2018, tariff announcements, tariff implementations, suspensions, retaliatory measures, and negotiation milestones created a sequence of observable information shocks. These shocks affected not only bilateral trade relations, but also expectations regarding global supply chains, corporate margins, consumer prices, and investor risk perception. Evidence from the 2018 tariff episode indicates that tariff measures affected U.S. prices and welfare [1], reinforcing the economic relevance of the conflict beyond political rhetoric. In addition, studies focused on firm exposure and trade linkages suggest that the financial market effects of the conflict were not uniform [4], as firms and sectors differed in their sensitivity to trade-related disruptions [6,7].

Accordingly, tariff-war events in this thesis are treated as historically dated information arrivals connected to observable market data. Financial markets react not only to realized policy changes, but also to expectations about future policy actions [2,3], so announcements, negotiations, suspensions, and implementation dates can all shift prices by altering beliefs about future costs, uncertainty, and macroeconomic conditions. The trade policy uncertainty literature supports this framing and provides the theoretical ba-

sis for comparing market reactions across event types, windows, and regimes [21].

2.1.3 Financial Market Reaction to Information Shocks

The analysis of tariff-war events also depends on the broader literature regarding how financial markets process information. The efficient market tradition suggests that asset prices tend to incorporate available information rapidly, especially when new information is public, relevant, and economically meaningful. The foundational work of Fama, Fisher, Jensen, and Roll demonstrated that stock prices adjust to new information, providing an important basis for later empirical studies of market reaction [9]. This view does not imply that markets are perfectly rational or that all reactions are immediate and complete, but it supports the use of market prices as observable indicators of how investors process information.

Market indices are useful in this context because they aggregate investor expectations across a broad set of firms. The S&P 500 does not directly measure trade flows, consumer welfare, or tariff revenues. Instead, it reflects how investors collectively reassess the value of large publicly traded U.S. firms when new information becomes available. This distinction is important for the interpretation of this thesis: the study does not estimate the full macroeconomic cost of the tariff war, but evaluates whether tariff-related information events were associated with abnormal movements in a broad U.S. equity benchmark.

In the context of policy shocks, financial market reactions may be influenced by both the content of the event and the degree to which the event was anticipated. Tariff-related information may affect expectations about future cash flows, production costs, supply chain disruptions, inflationary pressure, and geopolitical risk. As a result, market reactions may differ depending on whether an event represents escalation, relief, implementation, suspension, or agreement. Prior research on the U.S.–China trade conflict suggests that trade linkages and firm exposure can influence market value [4], supporting the relevance of event-level analysis [22] in this setting.

Tariff-war events in this thesis are accordingly interpreted as information shocks rather than isolated policy facts. The objective is not to prove that each event individually drove a specific market movement, but to examine whether abnormal market behavior is temporally associated with the arrival of tariff-related information. Geopolitical and trade-policy events unfold in complex macroeconomic environments where many forces move prices simultaneously, and the Event Study framework provides a structured method for evaluating associations while keeping interpretation anchored to what the data can actually support.

2.1.4 Event Study Methodology

The Event Study methodology is the central empirical framework of this research. Event studies are widely used in finance and economics to evaluate how asset prices

behave around specific information events. The method is grounded in the idea that, if markets incorporate information into prices, then economically meaningful events may be reflected in abnormal returns around the dates on which the information becomes public. The approach became prominent after the work of Fama, Fisher, Jensen, and Roll, who examined stock price adjustments to new information [9]. Brown and Warner later provided important methodological evidence on the use of daily stock returns in event studies, showing that event study techniques can perform effectively with daily return data when properly specified [10].

Mackinlay systematized the modern Event Study procedure by outlining the main steps of event definition, event window construction, expected return estimation, abnormal return calculation, and aggregation of abnormal returns [11]. Campbell, Lo, and Mackinlay also provide a broader econometric foundation for empirical financial market analysis, including the treatment of returns, risk, and statistical inference in financial data [12]. Together, these references provide the methodological foundation for the present thesis.

In practical terms, the Event Study framework compares observed returns during an event window with a benchmark of expected returns estimated from a pre-event period. The difference between observed and expected returns is defined as the abnormal return. By aggregating abnormal returns over a specified window, the researcher obtains the Cumulative Abnormal Return, which summarizes the total abnormal performance associated with the event window. In this thesis, the Event Study does not claim strict causality. Instead, it is used to evaluate whether tariff-war events are associated with market behavior that deviates from the expected pre-event baseline.

Methodological reviews also emphasize the importance of careful specification in event studies, particularly regarding event date selection, window length, expected return models, and statistical interpretation [23, 24]. These concerns are especially relevant in the present research because tariff-war events occurred during a broader period of economic uncertainty and later overlapped with the COVID-19 pandemic. As a result, the study distinguishes between a core tariff-war sample, an expanded market-covered sample, and broader documented events that are not necessarily used for financial inference.

2.1.5 Machine Learning Models Used in the Research

Two classification models are used in the exploratory Machine Learning layer: Logistic Regression and Random Forest. Both are standard tools in the supervised learning literature, and this section describes their design, the metrics used to evaluate them, and the precautions applied in a small, noisy financial dataset.

Logistic Regression classifies observations by fitting a linear decision boundary through a logistic (sigmoid) function, estimating class probabilities as a function of in-

put features. Its simplicity and interpretability make it a natural baseline in classification tasks and a useful reference against which more complex models can be compared [17, 18]. Here, it is used to predict the directional sign of market reaction from event-window features.

Random Forest, introduced by Breiman [16], builds an ensemble of decision trees, each trained on a bootstrap sample with a random feature subset at each node split, and combines predictions through majority voting. The ensemble design tends to reduce overfitting and handles nonlinear feature interactions more effectively than a single tree [17, 18]. It is included because the relationship between event-window features and market direction may not be linear, and because small-sample settings benefit from variance-reducing ensemble methods.

Model performance is evaluated by two complementary metrics. Accuracy captures the proportion of correctly classified observations, but in unbalanced datasets it can be misleading: a model that always predicts the majority class still achieves non-trivial accuracy without having learned anything. The F1-score, the harmonic mean of precision and recall, corrects for this by penalizing models that ignore minority classes [18].

A critical precaution in classification tasks is data leakage. In this thesis, the formal cumulative abnormal return (`formal_car_sp500`) is excluded from the feature set because it directly represents the target outcome. Including it would allow a model to learn the answer from the answer, producing inflated performance that carries no real predictive content.

Both models are evaluated against a majority-class baseline, which here equals 55.56%. A model failing to exceed this threshold has extracted no directional signal from the features. Because the 72 event-window observations are not fully independent (each underlying event contributes observations across three windows), the effective sample size is considerably smaller than the raw count implies. In noisy, limited data of this kind, model predictions should be scrutinized rather than taken at face value [19]. Machine Learning therefore functions in this thesis as an exploratory classification diagnostic: its role is to test whether the empirical patterns identified through abnormal returns translate into learnable directional signals, not to establish causal claims about tariff-war events.

2.2 RELATED WORKS

The following subsections survey the empirical literature most directly relevant to this thesis. The first subsection examines studies that measure financial-market reactions to U.S.–China tariff-war events using price-based and volatility-based methods, including Event Study designs and firm-level analyses. The second subsection reviews applied work on Machine Learning in financial event analysis, providing context for the exploratory classification layer used in this research. A comparative matrix summariz-

ing the methodological profile of each key study and the present thesis is presented at the end of this section.

2.2.1 Empirical Evidence on U.S.–China Tariff-War Market Reactions

Recent empirical studies on the United States–China tariff war suggest that financial-market reactions to trade-policy shocks are heterogeneous and depend on the level of analysis, the selected market channel, and the exposure of firms or sectors to international trade. This evidence is important for the present thesis because it indicates that aggregate index-level results may differ from firm-level, sector-level, volatility-based, or risk-premium-based findings.

Egger and Zhu analyze stock-market responses triggered by the United States–China trade war using an Event Study framework [25]. Their study examines how firms, sectors, countries, and global value-chain linkages reacted to tariff-war events. A central conclusion is that the effects of the trade war were not uniform. Tariffs affected directly targeted firms and sectors, but they also generated indirect effects through global production networks. Depending on sector, country, and supply-chain position, the observed stock-market responses could be positive or negative. This finding is relevant to the present thesis because it suggests that an aggregate index such as the S&P 500 may combine firms with different exposures, causing positive and negative reactions to partially offset each other in the average CAR.

Huang, Lin, Liu, and Tang examine the relationship between trade networks and firm value during the United States–China trade war [4]. Their study uses firm-level data and shows that market reactions to tariff announcements depend on direct and indirect exposure to trade linkages between the two economies. Firms more exposed to imports, exports, and global value-chain connections related to the conflict tend to experience more negative valuation effects and higher risk. This evidence reinforces the idea that tariff-war shocks may appear more clearly at the firm or sector level than in a broad aggregate index. Therefore, weak or statistically limited results in the S&P 500 do not necessarily imply that tariff-war events were irrelevant; rather, they may indicate that aggregate market indices mask internal heterogeneity across firms and sectors.

Ozdagli provides an important complementary perspective by examining the financial-market implications of the United States–China trade war through several market variables, including the S&P 500, VIX, high-yield spreads, and Treasury yields [26]. The study finds statistically and economically significant effects of trade-war events, especially through volatility, credit-spread, interest-rate, and equity-market channels. This result is important because it shows that different methodological choices and financial variables may detect effects that are not fully captured by a simple average CAR analysis of a broad equity index. For this reason, the findings of the present thesis should be interpreted as specific to the selected scope: daily S&P 500 event-window reactions,

rather than the full financial-market impact of the tariff war.

Positioned within this body of evidence, the present thesis asks whether tariff-war events produced a strong and systematic reaction in the aggregate daily S&P 500. The prior literature is clear that the financial consequences of the United States–China tariff war are real but heterogeneous, varying across firms, sectors, countries, value-chain exposure, volatility measures, and risk channels [4,25,26]. The cautious findings reported here are therefore not anomalous: market reactions to trade-policy shocks may be limited or difficult to detect at the aggregate index level, even when effects are present in more specific channels or more exposed units of analysis.

2.2.2 Data Science and Reproducibility in Empirical Finance

As empirical finance becomes increasingly computational, reproducibility has become a central methodological requirement. Financial studies often depend on multiple stages of data collection, cleaning, transformation, feature construction, model estimation, and visualization. Without clear documentation and verifiable workflows, results may become difficult to audit, replicate, or extend. Peng argues that replicable research is essential in computational science because it allows other researchers to understand how results were produced and to verify the analytical process [13]. Wilson et al. similarly emphasize practical standards for scientific computing, including organization, automation, documentation, and validation [14].

This thesis adopts these principles by organizing the empirical analysis as a modular research pipeline that is open to inspection. The project separates raw data, transformed features, event windows, abnormal returns, robustness outputs, figures, and validation scripts. This structure helps reduce the risk of hidden errors and makes each stage of the analysis easier to inspect. Although the study is not a software engineering project in the strict sense, it uses validation procedures inspired by the logic of Test-Driven Development. Beck’s work on Test-Driven Development provides the conceptual basis for incremental verification, which is adapted here as a practical strategy for checking empirical outputs rather than as a formal software development methodology [15].

The relevance of verifiability is particularly strong in this thesis because the analysis integrates several layers: event chronology, market data, event windows, abnormal returns, volatility, robustness, and exploratory Machine Learning. Each layer depends on previous outputs, meaning that an error in an early stage could propagate into later results. By validating intermediate datasets and preserving structured outputs, the research improves transparency and strengthens the credibility of the empirical procedure.

2.2.3 Machine Learning in Financial Event Analysis

Machine Learning has become increasingly common in financial analysis, especially

in tasks involving classification, prediction, pattern recognition, and feature importance. However, its use in small event-based financial datasets requires caution. Financial markets are noisy, and geopolitical events are difficult to encode into a limited number of numerical features. Because of this, the Machine Learning component in this thesis is not designed to replace the Event Study framework. Instead, it is used as an exploratory layer to test whether event-window features contain enough information to classify the directional sign of the market reaction.

The models used in this study are Logistic Regression and Random Forest. Logistic Regression is a classical supervised learning method widely used for binary classification because of its interpretability and simplicity. Random Forest, introduced by Breiman, is an ensemble learning method based on multiple decision trees and is commonly used when nonlinear relationships and interactions among features may exist [16]. Broader references in statistical learning also emphasize the importance of model evaluation, validation, and interpretability, especially when predictive models are applied to real-world data [17, 18]. Molnar's work on interpretable Machine Learning further supports the importance of understanding model behavior rather than treating predictions as black-box evidence [19].

In the context of this thesis, Machine Learning is interpreted conservatively. The target variable is based on the directional sign of the market reaction, and the formal CAR variable is excluded from the feature set to avoid direct leakage from the outcome into the predictors. The models are compared against a majority-class baseline, which is essential for evaluating whether the algorithms add detectable directional signal. If the models fail to outperform the baseline, this does not invalidate the Event Study. Instead, it indicates that the available event-window features may not contain sufficient predictive structure to classify market reactions reliably in a small sample.

Therefore, the Machine Learning layer contributes to the thesis as an exploratory classification diagnostic. It complements the Event Study by showing whether the empirical patterns identified through abnormal returns are easily captured by standard classification models. This distinction is important: the Event Study remains the core method for measuring market reactions around tariff-war events, while Machine Learning serves as a supplementary tool for examining the limits of prediction and classification in the constructed dataset.

2.2.4 Comparative Matrix of Related Works

Table 1 places the key studies surveyed above alongside the present thesis. The comparison covers the main methodological dimensions relevant to this research: use of Event Study design, measurement of cumulative abnormal returns, analysis of volatility or risk channels, robustness and sensitivity checks, exploratory Machine Learning, and a reproducible pipeline. The final column indicates how each study relates to the

methodological design of the present thesis.

Table 1: Comparative Matrix of Related Works

Study	Year	Research Focus	Event Study	CAR / AR	Volatility / Risk	Robustness	ML	Pipeline	Relation to This Thesis
Amiti, Redding & Weinstein	2019	Tariff welfare costs; firm pass-through	No	No	No	No	No	No	Economic basis; no event study
Baker, Bloom & Davis	2016	Trade Policy Uncertainty index	No	No	Partial	No	No	No	Motivates uncertainty framing
Caldara et al.	2020	TPU economic effects	No	No	Partial	No	No	No	Motivates regime separation
Egger & Zhu	2019	Firm/sector stock reactions to tariff war	Yes	Yes	No	No	No	No	Closest event study; firm-level
Huang et al.	2019	Firm value via trade-network exposure	Partial	No	Yes	No	No	No	Trade-network heterogeneity
Ozdagli	2019	VIX, spreads, equity channels	No	Partial	Yes	Partial	No	No	Multi-channel; motivates volatility layer
Podolski	2019	Trade policy and financial markets	Yes	Yes	No	No	No	No	Event study basis
Rao, Lucey & Kumar	2025	Tariff events; multi-country reactions	Yes	Yes	No	No	No	No	Escalation/relief design precedent
This thesis	2026	S&P 500 CAR; regime separation; ML layer	Yes	Yes	Yes	Yes	Yes	Yes	Full integration of all above

2.3 LITERATURE REVIEW CONCLUSION

The reviewed literature establishes that trade policy uncertainty significantly influences investor expectations [2, 3], firm valuations [4], and market volatility [26]. Empirical studies of the U.S.–China tariff war consistently caution, however, that these effects are conditioned by firm-specific exposure, sectoral nuances, global value-chain interconnectedness, and the choice of financial metrics, which are rarely uniform in direction or magnitude. At the aggregate level, S&P 500 outcomes may mask more granular dynamics accessible through firm- or sector-level analysis [4, 25, 26]. This heterogeneity is central to the present research design.

A recurring gap in the existing literature is the absence of studies that integrate a documented tariff-war chronology, regime-based separation, and rigorous robustness auditing alongside exploratory machine learning into a single replicable pipeline. This thesis addresses that gap by retaining the Event Study as the central empirical engine while augmenting it with disciplined data science practices and a carefully circumscribed machine learning layer.

The subsequent chapter translates this conceptual framework into a concrete empirical pipeline, detailing the operationalization of tariff-war events, market data processing, event-window construction, abnormal return estimation, and the implementation of robustness and classification protocols.



3

3

METHODOLOGY

3.1 RESEARCH DESIGN

This study adopts a quantitative, empirical, and applied research design to examine market reactions to key events within the United States–China tariff war on the U.S. equity market, with primary emphasis on the S&P 500. The central empirical framework is the Event Study methodology, which is appropriate for evaluating how financial markets react to discrete information shocks within predefined event windows [10–12].

The analysis does not purport to demonstrate econometric causality. It investigates temporal associations between known tariff-war events and subsequent market responses. This scope is important because financial markets are simultaneously shaped by monetary policy expectations, macroeconomic releases, geopolitical developments, and investor sentiment, none of which can be perfectly disentangled from trade-policy effects within this design. The Event Study methodology is appropriate precisely because it asks a narrower and more answerable question: whether returns around tariff-war events deviate systematically from the pre-event baseline.

To strengthen the empirical design, the study incorporates complementary analytical layers: volatility analysis, statistical testing, robustness and sensitivity checks, and an exploratory Machine Learning component. Each of these plays a defined supporting role: they qualify and stress-test the Event Study findings rather than replace them, by examining whether results hold across methodological specifications and whether event-window features carry enough directional signal to be classified by standard models.

3.2 GENERAL METHODOLOGICAL ARCHITECTURE

The methodological architecture of this research is structured as a single integrated pipeline, rather than as a collection of disconnected methods. The workflow begins with the definition of the research problem and the construction of a tariff-war event chronology. It then proceeds to regime separation, market data integration, financial feature construction, event window construction, expected return estimation, abnormal return calculation, CAR aggregation, volatility and signal analysis, statistical testing, robustness auditing, exploratory Machine Learning, validation, and final interpretation.

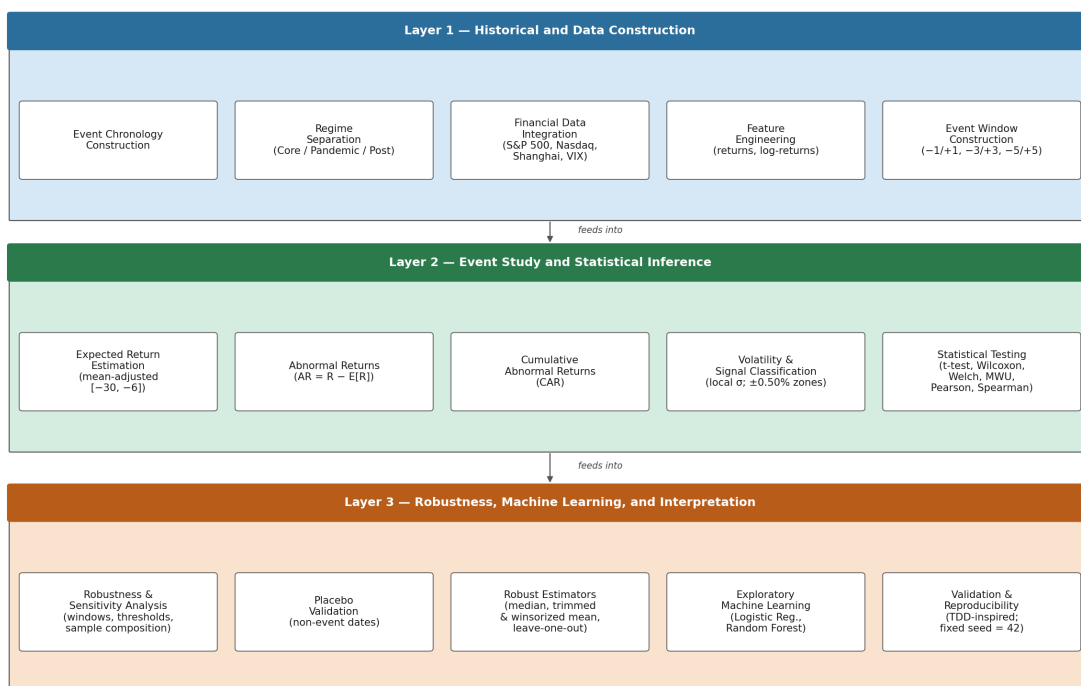


Figure 1: General Methodological Architecture of the Research

Figure 1 organizes the methodology into three sequential layers, each identified by a distinct color and feeding into the next. The first layer (in blue), *Historical and Data Construction*, establishes the empirical foundation of the study: it defines the tariff-war event chronology, separates the sample into the Core, Pandemic, and Post regimes, integrates market data from the S&P 500, Nasdaq, Shanghai, and VIX series, engineers the return and log-return features, and builds the symmetric event windows ($-1/+1$, $-3/+3$, $-5/+5$).

The second layer (in green), *Event Study and Statistical Inference*, constitutes the analytical core of the thesis. From the prepared data, it estimates expected returns through a mean-adjusted model over the $[-30, -6]$ window, derives abnormal returns as $AR = R - E[R]$, aggregates them into cumulative abnormal returns (CAR), classifies volatility and directional signals against the local σ and $\pm 0.50\%$ reference zones, and subjects the results to a battery of parametric and non-parametric tests (t-test, Wilcoxon, Welch, Mann–Whitney, Pearson, and Spearman).

The third layer (in orange), *Robustness, Machine Learning, and Interpretation*, stress-tests and extends these findings. It comprises sensitivity analyses over windows, thresholds, and sample composition; placebo validation on non-event dates; robust estimators (median, trimmed and winsorized means, leave-one-out); an exploratory Machine Learning stage (Logistic Regression and Random Forest); and validation and reproducibility procedures following a TDD-inspired design with a fixed seed of 42.

Read together, the diagram communicates that the Event Study is the analytical core of the thesis, while robustness analysis, volatility assessment, statistical testing, and Machine Learning operate as complementary layers. The validation and reproducibility procedures of the third layer support the entire pipeline rather than appearing only as a final step.

3.3 EVENT CHRONOLOGY CONSTRUCTION

The event chronology constitutes the empirical foundation of this research. The project developed a master chronology of 67 documented tariff-war events, organized into three temporal regimes: the core tariff-war phase, the pandemic period, and the post-pandemic period. This separation is methodologically important because the classical tariff-war period and the pandemic period are characterized by different macroeconomic conditions. Separating regimes helps prevent the analysis from treating distinct historical contexts as if they were a single homogeneous environment.

Within the master chronology, a refined core sample of 17 events was identified as representing the main information shocks of the classical U.S.–China tariff-war phase. In addition, 24 events had sufficient market coverage and were therefore included in the expanded financial inference; these are the events for which the main inferential tables in Chapter 4 are computed. Events without adequate market coverage remain documented in the master chronology for historical completeness and regime separation, but they are not used in the financial event study calculations.

The three-regime structure serves two purposes: historical documentation and methodological delimitation. The separation is justified by the fundamentally different macroeconomic contexts characterizing each phase. The core tariff-war phase (2018 to early 2020) unfolded in a relatively stable macroeconomic environment, with equity markets reacting to trade-policy announcements and negotiations in broadly normal conditions. The pandemic period (2020 to 2023) introduced an unprecedented disruption: global supply chains collapsed, equity volatility surged to historically exceptional levels, central banks undertook extraordinary monetary interventions, and investor behavior was dominated by health and fiscal uncertainty rather than by trade-policy signals alone. Treating these two phases as a single homogeneous environment would conflate market reactions driven by trade-policy announcements with reactions driven by pandemic-era systemic shocks. The post-pandemic period (2024 onward) again represents a distinct macroeconomic and political configuration, in which inflation dynamics, monetary policy normalization, and renewed geopolitical tensions define market conditions differently from both the classical tariff-war phase and the pandemic period.

By separating these regimes, the thesis avoids the distortion that would arise from averaging event-window returns across periods with dramatically different volatility baselines and market dynamics. The regime separation does not imply that no tariff-war

effects existed in the pandemic or post-pandemic phases. Rather, it acknowledges that identifying trade-policy-specific effects in those periods would require more specialized methods capable of controlling for the contemporaneous macro disturbances that dominated market movements. Within the scope of this thesis, the regime separation serves a documentation function and a scope-delimitation function: the core phase (2018–early 2020) and a subset of pandemic-period events (2020) contribute to the expanded market-covered sample ($N = 24$) used for financial inference, while post-pandemic events remain part of the historical chronology but lie outside the period of available daily financial data. Figure 2 illustrates the event classification and market-coverage filtering process described in this section.

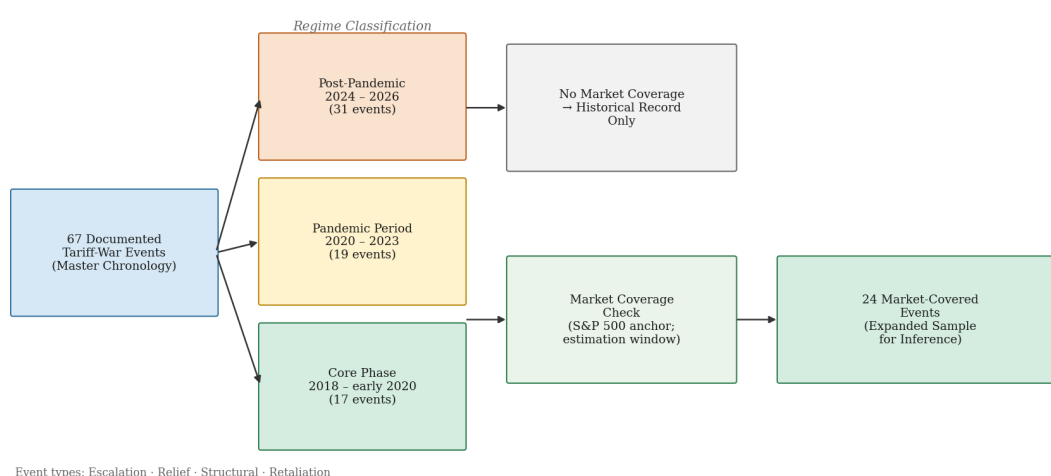


Figure 2: Event Chronology Construction: Regime Separation and Market-Coverage Filtering

Each event is categorized according to its trade-policy signal. Escalation events include tariff announcements, tariff implementations, tariff increases, and other actions that intensify trade tensions. Relief events include suspensions, reductions, agreements, negotiation progress, or actions that reduce or temporarily ease trade tensions. A third category, *structural*, is retained for events that modify the trade-policy framework without fitting neatly into escalation or relief. This classification is an analytical simplification designed to support repeatable comparison. It does not imply that each event is economically good or bad in an absolute sense; rather, it provides a structured way to compare how markets react to different directional trade-policy signals.

The master chronology also uses the label *retaliation* for policy responses that are reactive to prior actions by the other party. This label is retained as a historical descriptor for events outside the inferential sample (all market-covered events used in the Event Study are classified as escalation, relief, or structural) and is included in the appendix tables for historical completeness. Because no retaliation-labeled event appears in the 24 market-covered events that enter the financial analysis, this label does not affect

any reported inference.

For an event to be included in financial inference, it had to satisfy three operational requirements. First, the event needed a valid S&P 500 trading-day anchor date. Second, the available market data needed to cover the pre-event estimation window used to estimate expected returns. Third, the data needed to support the construction of the analyzed event windows. Events that did not satisfy these requirements were retained in the master chronology for historical completeness, but were excluded from the Event Study calculations. This criterion defines the meaning of sufficient market coverage in the thesis.

3.4 FINANCIAL DATA AND FEATURE CONSTRUCTION

The financial dataset is based on daily market data. The S&P 500 is used as the primary index because the research focuses on the U.S. equity market and seeks to evaluate how broad U.S. market prices reacted to tariff-war events. Additional indicators, including the Nasdaq Composite, the Shanghai Composite, and the VIX, are incorporated to provide complementary perspectives on technology-oriented firms, Chinese market conditions, and implied volatility expectations.

From the market price series, the project constructs daily returns and log-returns. These features provide the basis for expected return estimation, abnormal return calculation, volatility analysis, and exploratory Machine Learning. Missing values are treated conservatively and are not artificially fabricated. This is especially important because U.S. and Chinese markets follow different trading calendars, and holidays may produce missing observations in one market while another market remains open.

The market feature dataset used for financial inference covers the period for which reliable market observations are available. Events outside this market-covered period remain part of the broader historical chronology, but they are excluded from financial inference. This decision preserves methodological discipline by ensuring that every event used in the event study has adequate market data for window construction and return calculation.

3.5 EVENT WINDOW CONSTRUCTION

To capture immediate and short-term responses to tariff-war events, the study uses three symmetric event windows: $[-1, +1]$, $[-3, +3]$, and $[-5, +5]$ trading days. The $[-1, +1]$ window captures the immediate market reaction surrounding the event. The $[-3, +3]$ window captures a slightly broader short-term adjustment period. The $[-5, +5]$ window allows the analysis to observe whether the market reaction persists, attenuates, or partially reverses over a longer short-term horizon.

The windows are measured in trading days, not calendar days. For each event, the original event date is recorded as `event_date`. This date is then mapped to an

`anchor_date`, defined as the valid S&P 500 trading day used to anchor the event window. If an event occurred during a weekend or U.S. market holiday, the next valid S&P 500 trading day is used as the anchor. This procedure ensures that the analysis captures the first available market reaction after the event becomes observable to investors. Figure 3 illustrates the relationship between the estimation window, the anchor date, and the three event windows.

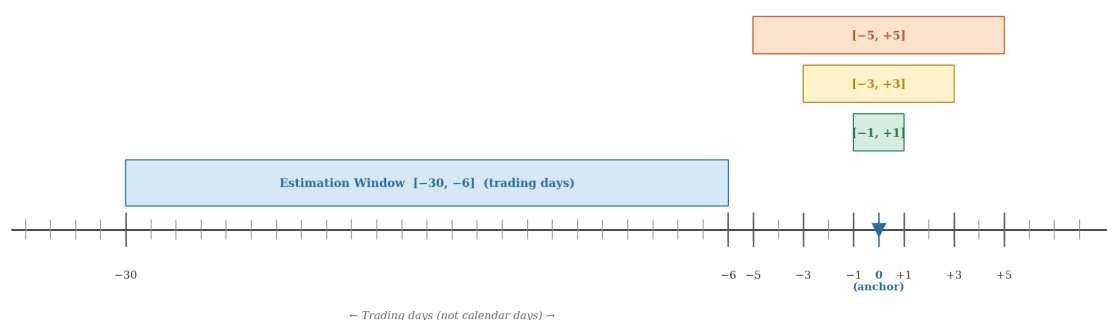


Figure 3: Event Window Construction: Estimation Window and Event Windows

3.6 EXPECTED RETURN ESTIMATION

The study uses a mean-adjusted return model to estimate normal market behavior prior to each event. The expected return is calculated using the pre-event estimation window $[-30, -6]$ trading days.

The choice of a pre-event estimation window follows the standard logic of Event Study methodology, as described by MacKinlay [11], Brown and Warner [10], and Campbell, Lo, and MacKinlay [12]: the expected return must be estimated from a period that precedes the event window and is therefore free from the price effects associated with the event itself. A pre-event estimation window provides a stable baseline while avoiding the contamination that would arise if the event window observations were included in the expected-return calculation.

The lower bound of -30 trading days provides a reasonably broad sample of pre-event daily returns without extending so far into the past that structural changes in market conditions or macroeconomic fundamentals distort the baseline. The upper bound of -6 is chosen to create a buffer between the end of the estimation window and the beginning of the nearest event window ($[-5, +5]$). This buffer is motivated by two concerns. First, it avoids including in the expected-return estimate the days immediately before the event, during which anticipatory price movements or pre-event information leakage may already have begun [23, 24]. Second, it ensures that the estimation window does not overlap with the widest event window used in the analysis. The $[-30, -6]$ choice is not claimed to be a universal standard; it is a transparent and defensible op-

erational decision for this study, compatible with the logic of Event Study estimation windows documented in the methodology literature. Its sensitivity is evaluated in the robustness analysis, which also tests estimation windows of $[-20, -6]$ and $[-40, -6]$ to verify that the main findings are not driven by this specific boundary choice.

The expected return for event i is defined as:

$$E(R_i) = \frac{1}{N_i} \sum_{t=-30}^{-6} R_{it}$$

where $E(R_i)$ is the expected return for event i , R_{it} is the observed daily return of the S&P 500 associated with event i on trading day t , and N_i is the number of valid observations available in the estimation window. The notation N_i is used instead of a fixed denominator to allow for possible missing observations in the financial dataset.

The mean-adjusted model is intentionally simple and transparent, but it also creates an important limitation that is especially relevant in this thesis. Because the primary object of analysis is the S&P 500 itself (a broad market index), the expected return is estimated from the S&P 500's own pre-event average return rather than from an external benchmark or a multifactor risk model. In this configuration, the abnormal return is the deviation of the S&P 500 from its own internal pre-event baseline; there is no separate market-wide control series absorbing broad market-wide shocks. Consequently, the estimated abnormal return may include both the market's specific response to the tariff-war event and any contemporaneous macro-financial disturbances that happen to coincide with the event window. This limitation is addressed through robustness checks, placebo validation, and cautious interpretation, but it remains a relevant boundary of the empirical design and is discussed further under Methodological Limitations.

3.7 ABNORMAL RETURNS

Abnormal returns measure the difference between the observed market return and the expected return estimated from the pre-event period. For each event i and trading day t within the event window, the abnormal return is defined as:

$$AR_{it} = R_{it} - E(R_i)$$

where AR_{it} is the abnormal return, R_{it} is the observed return, and $E(R_i)$ is the expected return estimated from the pre-event window. This calculation identifies the portion of the return that deviates from the pre-event baseline.

The logic follows MacKinlay's formulation of the Event Study method [11]: under the assumption that the estimation window provides an unbiased representation of normal market behavior, a positive AR_{it} indicates that the market returned more than expected on day t , and a negative AR_{it} indicates that it returned less than expected. These

deviations are interpreted as the market's response to information arriving in the event window. Because multiple factors can influence equity returns on any given day, they capture temporal association rather than structural causal effects [10, 23].

Figure 4 illustrates the conceptual structure of the abnormal return calculation.

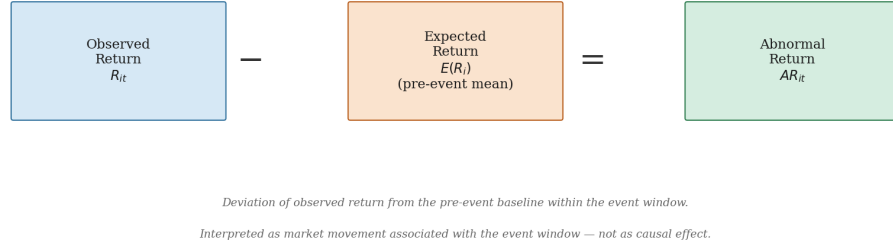


Figure 4: Abnormal Return: Observed Return Minus Expected Return

3.8 CUMULATIVE ABNORMAL RETURNS

Cumulative Abnormal Return is the main cumulative metric of the Event Study. It aggregates abnormal returns across a specific event window, allowing the analysis to summarize the total abnormal performance associated with that window. For event i and event window (τ_1, τ_2) , CAR is defined as:

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{it}$$

where $CAR_i(\tau_1, \tau_2)$ is the cumulative abnormal return for event i over the window from τ_1 to τ_2 , and AR_{it} is the abnormal return for event i on day t . A positive CAR indicates performance above the expected baseline during the event window, while a negative CAR indicates performance below the expected baseline.

Examining the sum of daily abnormal returns rather than a single day's return is important for several reasons. A single observation may be anomalous due to idiosyncratic factors that coincide with the event date. Aggregating abnormal returns over a multi-day window provides a more robust picture of the directional market movement associated with the event [11, 12]. At the same time, longer windows introduce more opportunity for unrelated events to contaminate the measurement. The three event windows used in this thesis $([-1, +1], [-3, +3], \text{ and } [-5, +5])$ are chosen to balance these concerns and to allow comparison of how the magnitude and direction of market reactions vary with window length. Comparing CAR values across windows also serves a robustness function: consistent negative or positive CAR patterns across all three windows provide stronger interpretive support than a result visible in only one window

specification.

The interpretation of CAR remains cautious throughout this thesis. A negative CAR does not necessarily mean that the tariff-war event caused a market decline; it means that the S&P 500 returned less during the event window than during the pre-event estimation period. Other contemporaneous developments may also have contributed to this deviation, as noted by Binder [23] and discussed in the methodological limitations section. Figure 5 provides an illustrative example of how daily abnormal returns accumulate into a CAR value.

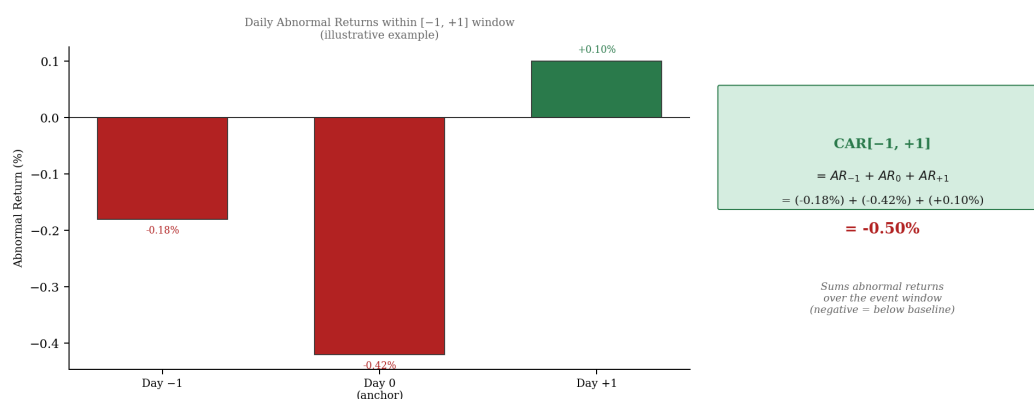


Figure 5: Cumulative Abnormal Return: Aggregation of Daily Abnormal Returns

3.9 VOLATILITY AND SIGNAL CLASSIFICATION

In addition to return-based measures, the study examines local volatility within event windows. Volatility is computed as the standard deviation of daily returns observed within the event window for each event. This measure captures the dispersion of returns around the event window, providing a proxy for the level of market instability associated with each tariff-war information shock. Examining volatility alongside CAR is important because a volatile market may produce large abnormal returns in either direction not because of a systematic directional response but because of general uncertainty or information ambiguity. Therefore, a period of high volatility surrounding an event weakens causal attributions even when the CAR itself is large in magnitude.

The study also classifies event-window outcomes into directional signal categories. A baseline neutral threshold of $\pm 0.50\%$ is used to distinguish small fluctuations from more directionally meaningful movements: outcomes with a CAR above $+0.50\%$ are classified as positive; outcomes below -0.50% as negative; outcomes in between as neutral. This categorical summary complements the quantitative CAR analysis by making patterns across event groups and windows easier to compare at a glance.

To assess the sensitivity of this classification to the choice of threshold, the robustness analysis also tests thresholds of $\pm 0.25\%$ and $\pm 0.75\%$. The main findings reported in the results chapter use the $\pm 0.50\%$ threshold. Figure 6 illustrates the three signal

zones and the classification logic.

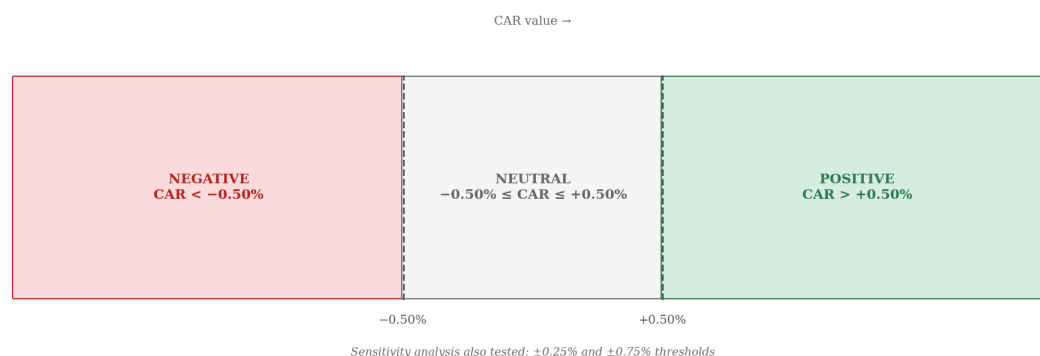


Figure 6: Signal Classification: Directional Zones by CAR Threshold

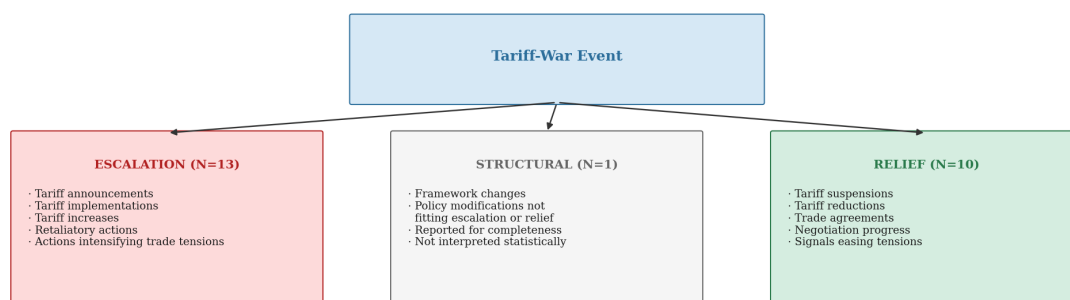
3.10 ESCALATION AND RELIEF EVENT GROUPS

The escalation and relief classification allows the study to compare whether markets respond differently to events that intensify trade tensions and events that reduce or temporarily ease them. Escalation events include tariff announcements, tariff increases, tariff implementations, retaliatory actions, and other measures that increase uncertainty or raise expected trade costs. Relief events include suspensions, tariff reductions, agreements, negotiation progress, and events that signal a reduction in trade tensions. A residual category, structural, is retained for events that modify the trade-policy framework without fitting clearly into either escalation or relief. The expanded market-covered sample includes 13 escalation events, 10 relief events, and 1 structural event. Because the structural group contains only a single market-covered observation, no inferential statistics are computed for it; it is reported for sample accounting completeness only.

This grouping supports the interpretation of asymmetric market responses. The central hypothesis motivating the comparison is that escalation events, by raising uncertainty and expected trade costs, would tend to produce negative market reactions, while relief events, by reducing uncertainty and expected costs, would tend to produce positive reactions. The empirical results reported in Chapter 4 allow this hypothesis to be evaluated systematically across the three event windows.

The classification is an analytical simplification designed to support repeatable, structured comparison. It is based on the direction of the trade-policy signal conveyed by each event, not on an assessment of its long-term welfare implications or economic desirability. Two events with opposite immediate trade-policy signals are classified into opposite groups even if their longer-run economic effects are ambiguous or similar. This approach is consistent with event study designs that group events by the informational signal they convey to investors rather than by their ultimate economic outcome [11,25].

Figure 7 illustrates the event classification logic.



*Classification is analytical, not a welfare or moral judgment.
Goal: compare short-window market reactions across policy signal directions.*

Figure 7: Escalation and Relief Event Groups: Classification Logic

3.11 STATISTICAL TESTING

The study applies conservative statistical tests suitable for a relatively small event sample. These tests are used to evaluate whether observed abnormal returns and CAR patterns provide suggestive evidence of systematic market reaction. The analysis includes comparisons across event windows, comparisons between escalation and relief groups, and correlation analysis between CAR and local volatility.

To assess whether the mean CAR across events is statistically distinguishable from zero, the study applies a one-sample t-test and a Wilcoxon signed-rank test. The one-sample t-test provides a parametric test based on the assumption of approximately normal CAR distributions within the sample, while the Wilcoxon signed-rank test is a nonparametric alternative that does not require this assumption and is more appropriate when sample sizes are small [24]. Bootstrap confidence intervals are also constructed through 5,000 resamples with a fixed seed of 42, providing an additional nonparametric assessment of the precision of mean and median CAR estimates.

To compare escalation and relief event groups, the study uses a Welch t-test, which does not assume equal variances across groups, and a Mann-Whitney U test, which is the nonparametric counterpart appropriate for comparing two independent samples without distributional assumptions. Using both a parametric and a nonparametric test for each research question increases robustness to assumptions about the underlying distribution of abnormal returns in a small sample.

Pearson and Spearman correlations are used to examine the association between cumulative abnormal returns and volatility. Pearson correlation captures linear association, while Spearman correlation captures rank-based monotonic association. Together, these provide a complementary picture of whether higher-volatility events tend to produce larger absolute abnormal returns.

All p-values are interpreted with caution. A lack of statistical significance does not prove that no effect exists, and statistical significance does not imply causality. The small sample size limits the statistical power of all tests, which means that non-significant results may reflect insufficient sample size rather than a genuine absence of systematic market reaction. This interpretation is consistent with the broader methodology literature on event studies with limited sample sizes [10, 11]. Figure 8 maps each research question to the corresponding statistical tests applied.

Research Question	Statistical Test(s)
Does CAR differ from zero?	One-sample t-test; Wilcoxon signed-rank; Bootstrap 95% CI (5,000 resamples, seed 42)
Do escalation and relief groups differ?	Welch t-test (unequal variances); Mann-Whitney U (non-parametric, two-sided)
Is CAR associated with local volatility?	Pearson correlation (linear); Spearman correlation (rank-based monotonic)
Are central tendency estimates robust to outliers?	Median, 10% trimmed mean, 10% winsorized mean; Sign test; Leave-one-out influence analysis

All tests interpreted with caution. Statistical significance alone does not imply causality.

Figure 8: Statistical Testing Design: Research Questions and Corresponding Tests

3.12 ROBUSTNESS AND SENSITIVITY ANALYSIS

Robustness and sensitivity analyses are used to assess the stability of the main interpretation under alternative methodological choices. The goal is not to prove that the findings are definitive, but to evaluate whether they are highly dependent on a single parameter setting.

The study tests three main robustness dimensions. First, alternative estimation windows are used: $[-20, -6]$, $[-30, -6]$, and $[-40, -6]$ trading days. Second, alternative signal thresholds are tested: 0.25%, 0.50%, and 0.75%. Third, sample composition is examined by comparing the core sample, the core-plus-pandemic-covered sample, and the expanded market-covered sample. Together, these tests help determine whether the central patterns remain reasonably stable across plausible analytical specifications.

Beyond the robustness checks regarding estimation windows, signal thresholds, and sample composition, the analysis also includes robust central-tendency diagnostics for the formal CAR estimates. Given the small size of the event sample, where individual observations might disproportionately skew the mean, the study compares the mean CAR with the median, the 10% trimmed mean, and the 10% winsorized mean across the three event windows. The analysis also applies a sign test and a leave-one-

out procedure to evaluate whether the aggregate CAR patterns are sensitive to individual events. These diagnostics do not replace the Event Study framework; they complement it by assessing whether the mean-based interpretation remains stable when less outlier-sensitive measures are considered.

3.13 EXPLORATORY MACHINE LEARNING LAYER

The Machine Learning layer is included as an exploratory complement to the Event Study framework. It uses Logistic Regression and Random Forest models to evaluate whether features derived from event windows contain enough directional signal to classify the sign of market reaction. Logistic Regression provides a simple and interpretable baseline for binary classification [17, 18], while Random Forest can capture nonlinear relationships [16] and interactions among features. These models are described conceptually in the literature review (see Subsection 2.1.5); this section focuses on their implementation in the context of this thesis. The models are evaluated against a majority-class baseline, and both accuracy and F1-score are reported to account for any class imbalance in the event-window dataset.

To reduce the risk of data leakage, `formal_car_sp500` is excluded from the feature set. This is necessary because the target variable is derived from the direction of market reaction, and including CAR directly as a predictor would allow the model to learn from information too closely related to the outcome. The excluded feature is a direct numerical representation of the cumulative market movement being predicted, meaning its inclusion would produce artificially inflated classification performance that has no predictive validity in practice. Figure 9 illustrates the feature set partitioning applied to prevent leakage.

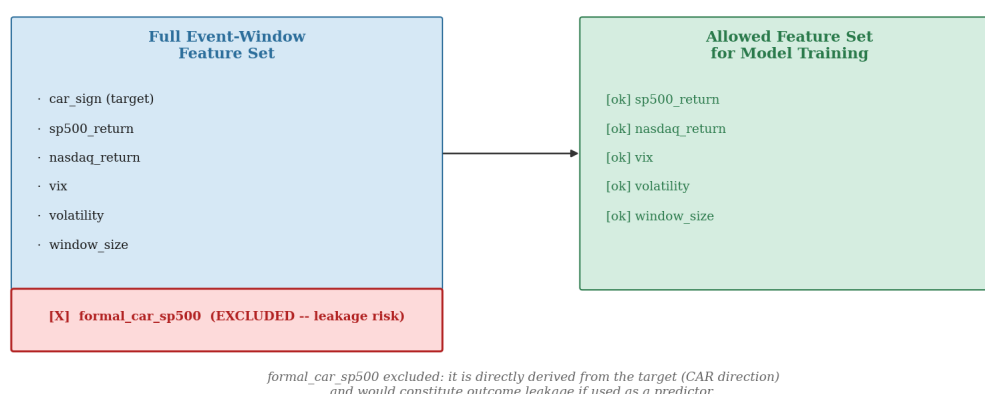
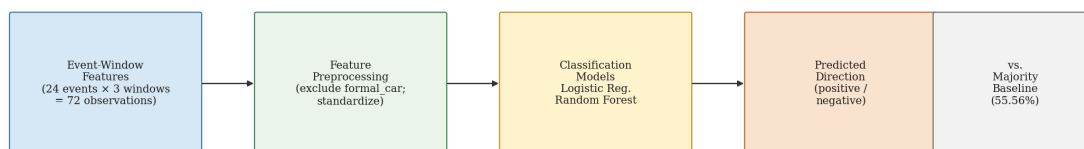


Figure 9: Data Leakage Prevention: Excluded Feature and Allowed Feature Set

The event-window structure creates an important limitation: multiple observations originate from the same underlying event because each event is represented across

three windows. Therefore, the Machine Learning dataset should not be interpreted as a large set of fully independent observations. The effective sample is much smaller than the raw observation count suggests, and this reinforces the need for conservative validation and cautious interpretation of all classification metrics. Figure 10 illustrates the full Machine Learning workflow from feature selection to model evaluation.



Models evaluated by accuracy and F1-score against majority baseline. Results interpreted as exploratory — not as primary evidence.

Figure 10: Exploratory Machine Learning Workflow: Features, Models, and Baseline Comparison

This layer is not presented as a causal model or as the primary empirical evidence of the thesis. It is a supplementary analysis designed to examine whether standard classification models can identify patterns in the constructed financial features. Interpretability and methodological caution are emphasized throughout the Machine Learning analysis [19].

3.14 VALIDATION, TDD, AND REPRODUCIBILITY

The project incorporates validation procedures inspired by Test-Driven Development [15] and by best practices in scientific computing [14] and reproducible computational research [13]. In this context, TDD is not used as a strict software engineering methodology. Instead, it serves as an inspiration for incremental validation: each stage of the empirical pipeline is checked before the next stage is built upon it.

The validation process addresses five categories of pipeline integrity. First, event chronology consistency checks verify that the master event list, regime labels, and event classifications are internally coherent. Second, market data coverage checks confirm that each event in the inferential sample has adequate pre-event and event-window observations and that no missing values appear where complete coverage is required. Third, event-window dataset checks verify that the generated feature matrices have the expected schema, observation counts, and value ranges. Fourth, output existence and figure validation checks confirm that all expected output files have been generated and that no prior analytical outputs have been inadvertently overwritten. Fifth, robustness output checks verify that alternative estimation windows, thresholds, and sample compositions are computed using the same pipeline logic as the main analysis.

Reproducibility is further strengthened by the use of a fixed random seed (`seed = 42`) for all stochastic operations, including bootstrap resampling and Machine Learning cross-validation. This ensures that every numerical result can be reproduced exactly by re-running the pipeline scripts in the documented order. Each validation script is named with the prefix `validate_` and is run after the corresponding construction script. A failed validation signals an inconsistency that must be resolved before downstream analysis proceeds, reducing the risk of silent errors propagating through the pipeline. Figure 11 illustrates the incremental validation logic.

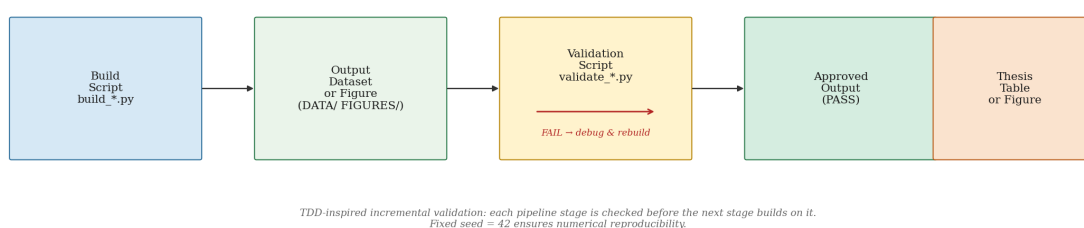


Figure 11: Validation Pipeline: Incremental Verification Across Pipeline Stages

3.15 COMPUTATIONAL IMPLEMENTATION

The empirical pipeline is implemented entirely in Python. The main dependencies are `pandas` and `numpy` for tabular data manipulation and numerical computation, `scipy` for statistical tests (Pearson, Spearman, Wilcoxon, Mann-Whitney, Binomial sign test, and bootstrap resampling), `scikit-learn` for the exploratory Machine Learning layer, and `matplotlib` for figure generation. Market data are accessed via `yfinance`.

The project is organized as a modular pipeline with explicit data-flow separation. Raw and intermediate datasets are stored in CSV format in the `DATA/` directory. Analysis scripts reside in `SCRIPTS/` and follow a naming convention that separates construction scripts (`build_*.py`), validation scripts (`validate_*.py`), and visualization scripts (`plot_*.py`). Figures are written to `FIGURES/` and are never modified manually. This structure ensures that any output can be traced back to a specific script and input file.

All random operations (bootstrap resampling and Machine Learning cross-validation) use a fixed random seed (`seed = 42`), making the numerical results deterministic and reproducible. The validation suite comprises multiple scripts that check each analytical layer against expected values, column schemas, row counts, and logical constraints. These validators are run after each build step to confirm that intermediate outputs are internally consistent and that no prior data have been inadvertently modified.

The project repository is publicly available at <https://github.com/joaopdmol/TCC---U.S.-CHINA-TARIFF-WAR-EVENTS>, where all scripts, datasets, and figures can be inspected

independently of the written thesis.

3.16 METHODOLOGICAL LIMITATIONS

The methodology has several limitations. First, the Event Study framework identifies temporal associations between tariff-war events and market reactions, but it does not establish strict econometric causality. Other macroeconomic, political, or financial developments may occur near the event dates and may influence market behavior.

Second, the analysis relies on daily market data. Daily data are appropriate for this thesis, but they do not capture intraday price discovery or immediate within-day reactions. Third, the event sample is relatively small, especially when divided across event groups, regimes, and windows. This limits the statistical power of the tests and requires cautious interpretation of p-values.

Fourth, and most specific to this thesis, the mean-adjusted return model is applied to the S&P 500, the same broad market index that serves as the object of analysis. This means the expected return is an *internal* pre-event benchmark rather than an external or risk-adjusted one. The resulting abnormal return measures how much the S&P 500 deviated from its own prior average during the event window; it does not fully remove broad market-wide shocks that are unrelated to the specific tariff-war event but that may coincide with the event date. The model is transparent and replicable, but it supports only a cautious interpretation: any observed abnormal return could partly reflect contemporaneous macroeconomic disturbances rather than a direct market response to the tariff event. This is the key conceptual reason why large outliers such as E08 (whose extreme CAR coincides with a period of broader late-2018 equity market stress) are not interpreted as evidence of a strong and isolated tariff-war reaction, and why the thesis relies on medians, trimmed means, placebo tests, and leave-one-out diagnostics. Future research should compare these results against alternative expected-return specifications, such as a market model using an external benchmark, the CAPM, or Fama-French factors, to evaluate whether the estimated CARs are robust to the choice of baseline.

3.17 METHODOLOGICAL SYNTHESIS

This thesis is anchored in the Event Study framework. Its core empirical task is to compare observed S&P 500 returns around tariff-war events against a pre-event benchmark and then summarize those deviations through abnormal returns and Cumulative Abnormal Returns. The additional analytical layers are introduced to qualify this evidence rather than to replace it.

The event chronology establishes the historical object of analysis, while the market-covered sample sets the boundaries of financial inference. Keeping these two scopes separate matters because the thesis documents a broader sequence of tariff-war events,

yet only those with valid market coverage can support return-based calculations. In practical terms, the methodology distinguishes what can be historically documented from what can be empirically measured through daily financial data.

Each supporting component (the robustness checks, placebo test, volatility analysis, and exploratory Machine Learning layer) plays a clearly defined role. The robustness checks test whether the results hinge on a single methodological choice. The placebo test asks whether the observed CAR patterns look unusual when measured against randomly selected non-event dates. Volatility analysis adds an extra layer of information about market instability around event windows. The Machine Learning layer, in turn, is treated only as a diagnostic of directional predictability under small-sample conditions.

The value of this design lies precisely in what it makes explicit: which patterns are window-sensitive, which estimates are driven by individual events, and where the evidence stops short of supporting causal inference. The Event Study remains central because it answers the primary empirical question; the additional layers define the boundaries within which that answer is credible.



4

4

RESULTS

4.1 DESCRIPTIVE OVERVIEW OF EVENTS AND MARKET DATA

The master chronology comprises 67 identified tariff-war events, categorized according to their classification as escalation or relief actions and organized across different historical regimes. Within this broader chronology, the core Event Study sample focuses on 17 primary events associated with the classical phase of the United States–China tariff war. The expanded market-covered analysis includes 24 events with sufficient market data for financial inference. In addition, the dataset used for the exploratory Machine Learning layer contains 72 event-window observations, reflecting the combination of covered events and event windows.

Table 2 summarizes the empirical structure used in the thesis. This matters because the project contains a broad historical chronology, but financial inference is restricted to events with sufficient market data. Therefore, the master chronology provides historical coverage, while the core and expanded market-covered samples define the scope of the Event Study and the exploratory Machine Learning dataset.

Table 2: Composition of the Empirical Samples

Component	Count
Master event chronology	67
Core Event Study sample	17
Expanded market-covered sample	24
Machine Learning event-window observations	72

Descriptive inspection of the market data reveals periods of localized volatility around several tariff-related events, particularly during phases of intense trade tension. Figure 12 positions the core tariff-war events along the S&P 500 trajectory, offering a visual overview of how escalation and relief events were distributed across the classical phase of the conflict.

The visual pattern confirms the expected concentration of events in 2018–2019, with a visible spike in market volatility during the most intense phase of the escalation.

4.2 MAIN EVENT STUDY RESULTS

S&P 500 and core U.S.-China tariff war timeline (2017–2020)

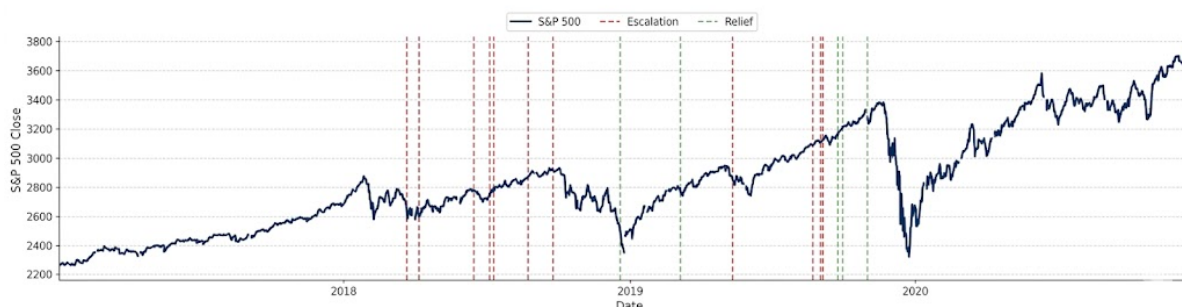


Figure 12: S&P 500 and Core U.S.–China Tariff War Timeline

The main Event Study results indicate a negative short-window reaction followed by attenuation in longer windows. For the expanded market-covered sample, the aggregate cumulative abnormal returns are approximately -0.23% in the $[-1, +1]$ window, -0.16% in the $[-3, +3]$ window, and 0.11% in the $[-5, +5]$ window. These values suggest that the immediate market response to tariff-war events was negative on average, while the effect became less negative or slightly positive as the event window expanded.

Table 3 consolidates the average formal CAR values across the three event windows used in the analysis. Presenting these values in tabular form makes the temporal pattern clearer and allows the reader to compare the immediate and short-term reactions to tariff-war events.

Table 3: Average Formal CAR by Event Window

Event Window	Average Formal CAR
$[-1, +1]$	-0.23%
$[-3, +3]$	-0.16%
$[-5, +5]$	0.11%

Table 3 shows that the average formal CAR is negative in the shorter windows and becomes slightly positive in the $[-5, +5]$ window, tracing an initial adverse reaction that attenuates or partially reverses as the horizon expands. Comparing multiple windows is therefore not merely an exercise in sensitivity testing: the sign of the aggregate reaction itself changes depending on the chosen horizon, which is a substantive finding about the temporal structure of market responses to tariff-war events.

Short windows tend to capture the immediate market response to new information, while longer windows allow for adjustment, correction, or partial reversal [10, 11]. Against the null hypothesis of zero abnormal return, however, neither the window-level nor the group-level tests reach statistical significance at the 5% level. The Event Study findings are therefore best read as evidence of temporal association and market sen-

sitivity, not as proof that each tariff-war event individually caused the observed price movement.

The correlation analysis between CAR and local volatility adds texture to this reading. Abnormal market reactions tend to be larger in absolute magnitude during periods of higher local volatility around event windows, suggesting that tariff shocks may interact with broader market instability rather than driving volatility independently.

To make the inferential interpretation more transparent, Table 4 reports the numerical evidence behind the Event Study results. The table summarizes the number of event-window observations, mean CAR, median CAR, standard deviation, parametric and non-parametric p-values, and bootstrap confidence intervals for each analyzed window. This structure allows the reader to verify directly why the results are interpreted cautiously.

Table 4: Inferential Summary of Formal CAR by Event Window

Window	N	Mean CAR	Median CAR	SD	t-test p	Wilcoxon p	Bootstrap 95% CI
$[-1, +1]$	24	-0.23%	-0.06%	1.65%	0.494	0.812	$[-0.92\%, 0.39\%]$
$[-3, +3]$	24	-0.16%	0.48%	3.36%	0.813	0.623	$[-1.59\%, 1.06\%]$
$[-5, +5]$	24	0.11%	1.06%	3.40%	0.878	0.833	$[-1.27\%, 1.41\%]$

Notes: CAR refers to the formal cumulative abnormal return of the S&P 500. The t-test evaluates whether the mean CAR differs from zero. The Wilcoxon signed-rank test evaluates whether the median CAR differs from zero when technically feasible. Bootstrap confidence intervals are based on 5,000 resamples of the mean with seed 42.

Table 4 shows that the estimated CAR values are small in magnitude and statistically weak across all three event windows. None of the reported p-values reach the 5% threshold, and the bootstrap confidence intervals include zero in every specification. These two facts together are the empirical floor: however the results are parsed (by window, by group, or by estimator), no specification yields strong evidence of a systematic or causal market reaction. The aggregate pattern instead suggests non-exceptional abnormal returns relative to the pre-event baseline.

One event deserves specific attention in the interpretation of the aggregate results. Event E08, associated with the postponement of a scheduled tariff increase on 2018-12-19, produced a formal CAR of -11.02% in the $[-3, +3]$ window, making it one of the most negative event-level observations in the sample. This value is reported in the event-level audit table in the appendix and should be treated as an important outlier. Its magnitude suggests that the aggregate results, especially group averages for relief events, may be influenced by broader market stress occurring around that date.

This does not invalidate the Event Study, but it limits the strength of group-level interpretation. The E08 window coincided with broader market stress in late 2018. Because this thesis applies a mean-adjusted model to the S&P 500 itself, the abnormal return for E08 measures the deviation of the S&P 500 from its own pre-event average: there is no external benchmark absorbing broad market-wide movements. As a result,

the model cannot separate the component of E08's CAR that reflects the specific tariff postponement from the component that reflects the simultaneous macro-financial turbulence in Q4 2018. E08 therefore illustrates not only why the thesis uses medians, trimmed means, placebo tests, and leave-one-out diagnostics, but also why the mean-adjusted design supports cautious temporal-association language rather than causal claims about individual events.

4.3 ESCALATION VS RELIEF EVENTS

The comparison between escalation and relief events reveals that market responses were not symmetric across event types. Escalation events sometimes carry higher average CARs than relief events in certain windows, a pattern that should not be read as evidence of economic benefit. A more defensible interpretation is that market reactions depend on expectations, prior pricing, uncertainty resolution, and the perceived informational content of each event rather than on its policy direction alone.

Some escalation events may resolve a period of uncertainty by clarifying the policy path, even when the policy itself is negative. When investors had partially anticipated the escalation, the announcement may function more as confirmation than as surprise. By the same logic, relief events may generate weaker abnormal returns if negotiations, postponements, or partial agreements had already been priced in before the formal announcement date. Escalation and relief are therefore best understood as analytical categories for comparing short-window reactions, not as direct economic labels.

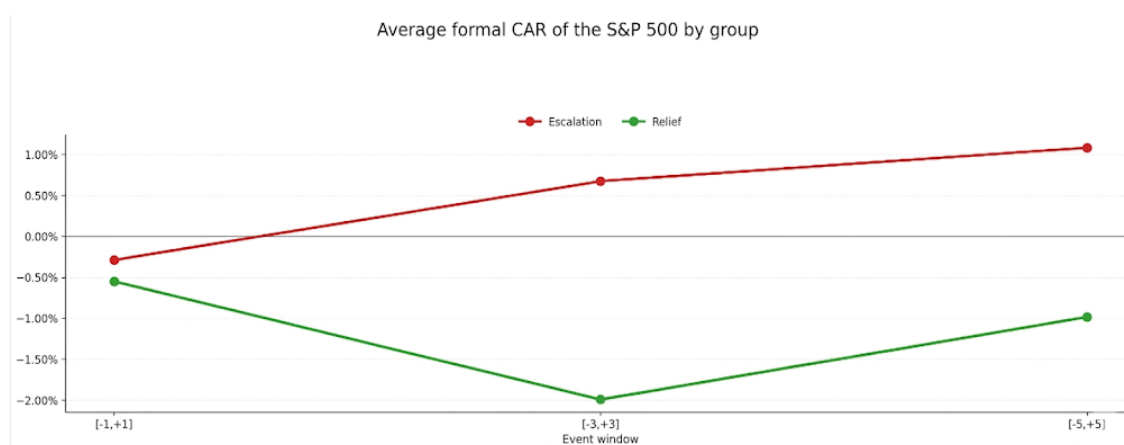


Figure 13: Average Formal CAR of the S&P 500 by Event Group

Figure 13 compares the average formal CAR of escalation and relief events across the analyzed event windows. The visual pattern suggests that market reactions were not symmetric across event groups, reinforcing the interpretation that the informational content of tariff-war events depends on expectations, timing, and the perceived direction of trade-policy signals.

To complement the visual comparison, Table 5 reports the formal CAR values by

event group and event window. This table separates escalation, relief, and structural events in order to show how the descriptive CAR pattern changes across the short event horizons used in the Event Study.

Table 5: Formal CAR by Event Group and Window

Group	Window	N	Mean CAR	Median CAR	SD
Escalation	$[-1, +1]$	13	-0.31%	-0.22%	1.96%
Escalation	$[-3, +3]$	13	0.48%	1.15%	3.08%
Escalation	$[-5, +5]$	13	0.96%	1.69%	3.32%
Relief	$[-1, +1]$	10	-0.13%	0.16%	1.35%
Relief	$[-3, +3]$	10	-0.98%	0.48%	3.85%
Relief	$[-5, +5]$	10	-0.92%	0.74%	3.54%
Structural	$[-1, +1]$	1	-0.35%	-0.35%	<i>n/a</i>
Structural	$[-3, +3]$	1	-0.38%	-0.38%	<i>n/a</i>
Structural	$[-5, +5]$	1	-0.66%	-0.66%	<i>n/a</i>

The formal CAR in Table 5 refers to the cumulative abnormal return of the S&P 500, grouped by event classification and event window. The Structural category contains one market-covered event and is reported only for sample accounting completeness; it is not interpreted statistically because no inference is possible with a single observation. The standard deviation is reported as *n/a* for this group because it is not defined for a single observation.

Table 5 indicates that escalation and relief events followed different descriptive patterns across the analyzed windows. Escalation events shift from a negative average CAR in the immediate $[-1, +1]$ window to positive average CAR values in the wider windows, while relief events remain negative on average across all three windows despite positive median values. The Structural category is reported only for sample accounting completeness and is not interpreted statistically. Overall, the table supports a descriptive asymmetry across event classifications, but it does not establish causal effects or statistically definitive group differences.

To evaluate whether the observed descriptive differences between escalation and relief events are statistically meaningful, Table 6 reports Welch *t*-tests and Mann-Whitney *U* tests comparing the two groups, using the expanded market-covered sample (escalation $N = 13$, relief $N = 10$).

As shown in Table 6, neither the Welch *t*-test nor the Mann-Whitney *U* test reaches statistical significance in any event window. The descriptive mean differences visible in Table 5 (particularly the divergence in the $[-3, +3]$ and $[-5, +5]$ windows) remain unconfirmed at conventional thresholds. The group-level asymmetry is therefore better attributed to the heterogeneous informational content of individual events than to a systematic difference in how markets process escalation versus relief signals.

Table 6: Formal Comparison of Escalation and Relief Event Groups

Window	Mean Esc.	Mean Rel.	Welch t p	MW U p	Sig. (5%)
$[-1, +1]$	-0.31%	-0.13%	0.802	0.828	No
$[-3, +3]$	0.48%	-0.98%	0.337	0.369	No
$[-5, +5]$	0.96%	-0.92%	0.211	0.227	No

Notes: Escalation $N = 13$, relief $N = 10$ (expanded market-covered sample; structural event excluded). Welch t -test assumes unequal variances. Mann-Whitney U uses a two-sided alternative. None of the tests reach the 5% significance level, confirming that the observed group differences are descriptive rather than statistically established.

4.4 WINDOW-BASED COMPARISON

The three event windows ($[-1, +1]$, $[-3, +3]$, and $[-5, +5]$) trace a clear temporal gradient in the cumulative abnormal returns. The shortest window, where the average CAR is negative, captures the immediate market response to tariff-related information. Table 4 shows how that negative reaction attenuates across horizons: by the $[-5, +5]$ window, the average CAR for the expanded market-covered sample has shifted to slightly positive. Whether this shift reflects partial absorption, correction, or reassessment is difficult to establish, but it confirms that the chosen event horizon is not a neutral technical choice: it shapes the substantive interpretation of whether the market reacted negatively or positively to tariff-war events.

4.5 VOLATILITY AND SIGNAL DISTRIBUTION

The volatility analysis provides descriptive insight into market behavior surrounding tariff-war events. Local volatility is used to evaluate whether event windows with stronger abnormal returns are also associated with more unstable market conditions.

Table 7 reports Pearson and Spearman correlations between the formal CAR and local S&P 500 volatility, computed on the expanded market-covered sample ($N = 24$ events, 72 pooled event-window observations).

Table 7: Correlation Between Formal CAR and Local Volatility (Expanded Sample, $N = 24$)

Scope	Obs.	Pearson r	Pearson p	Spearman r	Spearman p
Pooled (all windows)	72	-0.318	0.007	-0.170	0.153
Window $[-1, +1]$	24	-0.335	0.110	-0.143	0.506
Window $[-3, +3]$	24	-0.220	0.302	-0.050	0.818
Window $[-5, +5]$	24	-0.452	0.027	-0.300	0.154

Notes: Formal CAR is the cumulative abnormal return of the S&P 500. Local volatility is the within-window standard deviation of daily S&P 500 returns. The pooled analysis combines all 72 event-window observations (24 events $\times$ 3 windows); standard errors are not corrected for within-event clustering.

The results reveal a weak negative association: Pearson correlation is -0.318 in the pooled sample ($p = 0.007$) and -0.452 in the $[-5, +5]$ window ($p = 0.027$). However, the rank-based Spearman correlation is not statistically significant in any specification, which suggests that the Pearson result is sensitive to the influence of a small number of observations with extreme values, consistent with the outsized role of E08 documented in the main Event Study results. Within individual windows, neither Pearson nor Spearman reaches conventional significance at the 5% level for the $[-1, +1]$ and $[-3, +3]$ windows. Therefore, the evidence for a systematic association between CAR and local volatility is weak and not robust across correlation methods. Rather than indicating that higher volatility drives negative CAR in a predictable way, the negative sign suggests only that the most volatile event windows tend to be those with the most negative abnormal returns, a pattern that is consistent with, but not causally established by, the presence of E08.

The signal distribution analysis classifies event-window outcomes as positive, neutral, or negative based on the formal CAR and the predefined neutral threshold. This classification does not replace the CAR analysis. Instead, it provides a compact way to summarize how market reactions are distributed across event windows and event groups. The distribution of signals helps clarify whether the results are driven by a small number of extreme events or whether positive and negative responses are more broadly spread across the sample.

4.6 CORE VS EXPANDED SAMPLE

The comparison between the core sample and the expanded market-covered sample is important for evaluating whether the findings depend on the original 17-event core selection. The master chronology contains 67 documented events, but not all of them are used for financial inference because some fall outside the available market-covered period. Therefore, the relevant robustness comparison is between the core sample and the expanded sample with sufficient market data.

The results indicate that the market response appears more pronounced during the core phase of the tariff war. When the expanded market-covered sample is included, the magnitude of the results tends to attenuate. This attenuation suggests that the core phase may contain more concentrated tariff-war shocks, while the expanded covered sample incorporates a broader and potentially more heterogeneous set of events. Importantly, the attenuation does not necessarily reverse the main interpretation: tariff-war events remain temporally associated with measurable market reactions, but the magnitude and direction of those reactions depend on sample composition and event window.

Figure 14 compares the core sample with the expanded market-covered sample. The attenuation observed in the expanded sample suggests that the core phase con-

Comparison of Average Formal CAR by Window: Core vs. Expanded

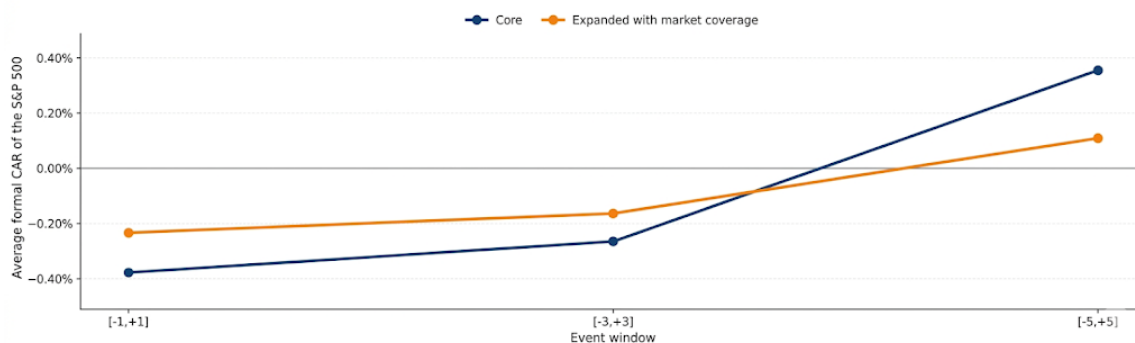


Figure 14: Core Versus Expanded Market-Covered Comparison

centrated stronger tariff-war shocks, while the expanded market-covered sample incorporated a broader and more heterogeneous set of events. This supports the decision to distinguish historical documentation from financial inference.

Table 8 presents the numerical comparison between the two samples across the three event windows. The main inferential tables in this thesis (Tables 4, 10, and 11) all report results for the expanded market-covered sample ($N = 24$). The core sample ($N = 17$) is retained as a robustness reference because it represents the classical tariff-war phase and can be compared directly against the expanded sample.

Table 8: Core Sample vs. Expanded Market-Covered Sample: Formal CAR Comparison

Window	N (Core)	Mean (Core)	Median (Core)	N (Exp.)	Mean (Exp.)	Median (Exp.)
$[-1, +1]$	17	-0.38%	0.22%	24	-0.23%	-0.06%
$[-3, +3]$	17	-0.27%	1.10%	24	-0.16%	0.48%
$[-5, +5]$	17	0.35%	1.36%	24	0.11%	1.06%

Notes: Core sample ($N = 17$) consists of the classical tariff-war events from 2018–2020. Expanded market-covered sample ($N = 24$) adds seven additional events from the pandemic period with sufficient market data. The mean and median in both samples are descriptive, without inferential adjustments for sample-selection effects.

The comparison confirms the attenuation pattern: mean and median CAR values are slightly weaker in the expanded sample than in the core sample across the $[-1, +1]$ and $[-3, +3]$ windows, with similar attenuation in the $[-5, +5]$ window. Crucially, the directional pattern holds across both samples (negative in short windows, tending toward or above zero in the widest window), which supports the robustness of the main interpretation without requiring that both samples tell the identical story. The main inferential tables in this chapter use the expanded sample ($N = 24$) as the primary analytical frame because it makes fuller use of the available market data.

4.7 ROBUSTNESS AND SENSITIVITY RESULTS

The robustness and sensitivity analysis evaluates whether the main interpretation holds under alternative methodological choices, varying the estimation window, the signal classification threshold, and the sample composition. The question is not whether results stay exactly constant (they do not) but whether they shift so sharply under plausible alternatives that the central interpretation becomes untenable.

Across specifications, the direction of the results is broadly stable. The shorter $[-20, -6]$ estimation window produces some changes in average CAR magnitude, while the longer $[-40, -6]$ window remains closer to the baseline. The findings are therefore not mechanical artifacts of one estimation window, though the precise magnitudes vary enough that they should not be taken at face value. Robustness here means stability of direction under reasonable alternatives, not invariance of magnitude across all specifications.

Table 9 evaluates the sensitivity of signal classification to the choice of neutral threshold. For each event window and each threshold ($\pm 0.25\%$, $\pm 0.50\%$, $\pm 0.75\%$), the table reports the number of events classified as positive, neutral, and negative in the expanded sample ($N = 24$).

Table 9: Signal Classification Sensitivity to Neutral Threshold (Expanded Sample, $N = 24$)

Window	Threshold	Positive	Neutral	Negative
$[-1, +1]$	$\pm 0.25\%$	9	4	11
$[-1, +1]$	$\pm 0.50\%$	8	7	9
$[-1, +1]$	$\pm 0.75\%$	6	13	5
$[-3, +3]$	$\pm 0.25\%$	13	2	9
$[-3, +3]$	$\pm 0.50\%$	12	5	7
$[-3, +3]$	$\pm 0.75\%$	11	6	7
$[-5, +5]$	$\pm 0.25\%$	14	1	9
$[-5, +5]$	$\pm 0.50\%$	13	2	9
$[-5, +5]$	$\pm 0.75\%$	13	3	8

Notes: Positive denotes $CAR > +\text{threshold}$; negative denotes $CAR < -\text{threshold}$; neutral denotes $|CAR| \leq \text{threshold}$. The baseline threshold used throughout the thesis is $\pm 0.50\%$.

The signal classification is broadly stable across threshold choices in the $[-3, +3]$ and $[-5, +5]$ windows, where positive signals consistently outnumber negative signals regardless of the threshold. The $[-1, +1]$ window is the most sensitive to the threshold: at $\pm 0.25\%$ negative signals dominate, while at $\pm 0.75\%$ the majority of events fall within the neutral zone. This sensitivity reflects the small average CAR magnitude in the immediate window and reinforces the interpretation that short-window results should not be read as evidence of a systematic directional response.

Figure 15 reports how average CAR changes when alternative estimation windows are used. The figure supports the robustness discussion by showing whether the main

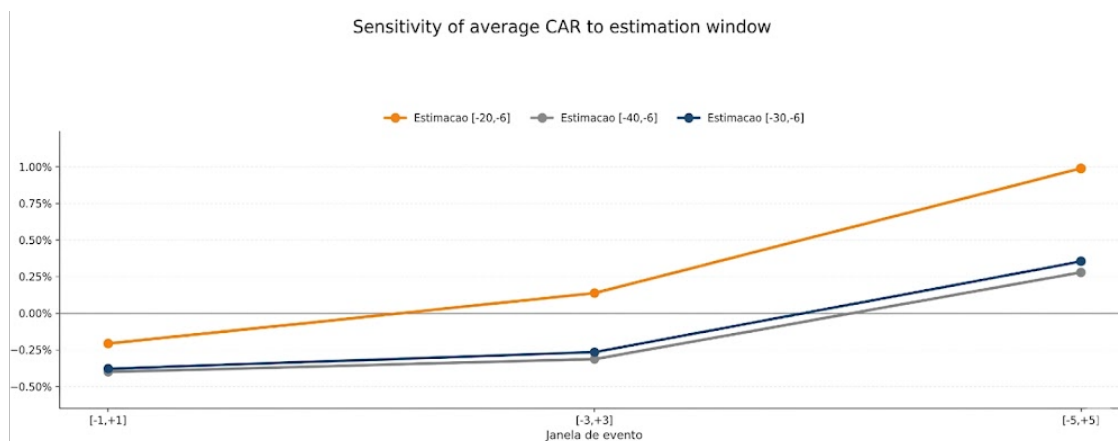


Figure 15: Sensitivity of Average CAR to Estimation Window

interpretation is highly dependent on the baseline $[-30, -6]$ estimation window or remains reasonably stable under alternative pre-event windows.

Table 10: Robust Central-Tendency Estimators of Formal CAR by Event Window

Event Window	N	Mean CAR	Median CAR	10% Trimmed Mean	10% Winsorized Mean
$[-1, +1]$	24	-0.23%	-0.06%	-0.10%	-0.19%
$[-3, +3]$	24	-0.16%	0.48%	0.19%	0.11%
$[-5, +5]$	24	0.11%	1.06%	0.21%	0.07%

Notes: CAR refers to the formal cumulative abnormal return of the S&P 500. Mean and median are reproduced from the inferential summary for direct comparison with robust estimators. Trimmed and winsorized means use a 10% fraction at each tail.

As an additional robustness layer, Table 10 compares the mean CAR with less outlier-sensitive central-tendency estimators. In the immediate $[-1, +1]$ window, all estimators remain negative or close to zero. From the $[-3, +3]$ window onward, however, the median, trimmed mean, and winsorized mean become positive while the simple mean remains slightly negative. This indicates that the negative mean in the $[-3, +3]$ window is influenced by a small number of strongly negative observations rather than by a broad-based negative market reaction.

The one-sample sign test reinforces this cautious interpretation. The sign-test p -values are 1.000, 0.541, and 0.541 for the $[-1, +1]$, $[-3, +3]$, and $[-5, +5]$ windows, respectively. Therefore, the null hypothesis of a zero median cannot be rejected in any event window.

The leave-one-out analysis further clarifies the role of individual observations in the aggregate mean. Event E08, already discussed in the main Event Study results for its -11.02% CAR in the $[-3, +3]$ window, is the most influential observation in that specification. Excluding E08 shifts the mean from -0.16% to +0.31%, a change of +0.47 percentage points. E08 is therefore retained in the sample, while the diagnostic is used to document its influence and to clarify why the mean diverges from the robust estimators in that window.

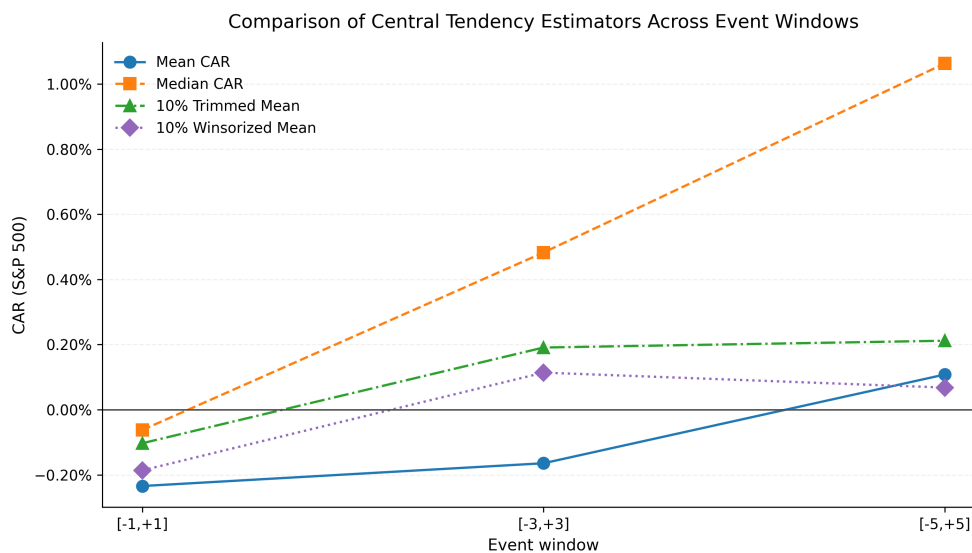


Figure 16: Comparison of Mean, Median, Trimmed and Winsorized Formal CAR by Window

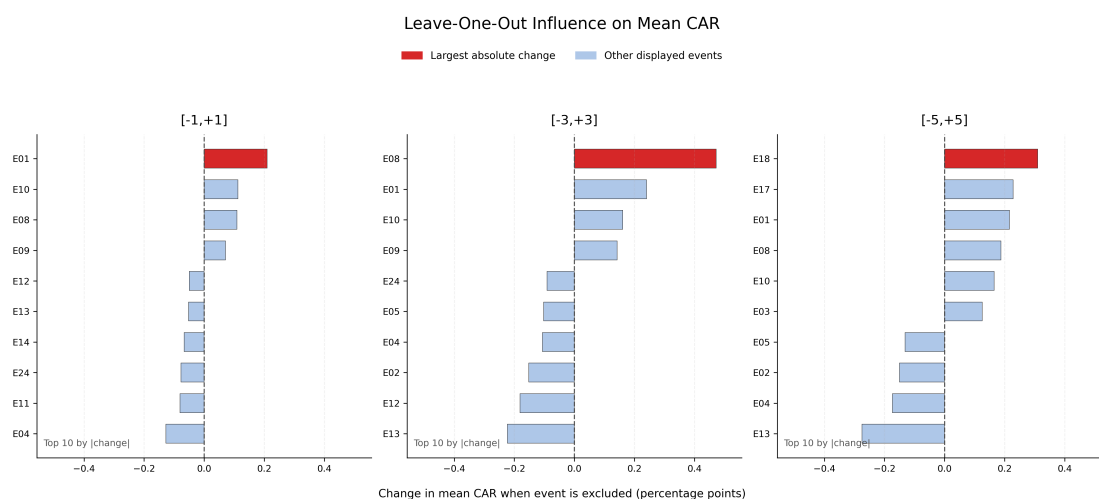


Figure 17: Leave-One-Out Influence of Individual Events on Mean Formal CAR

4.8 PLACEBO ROBUSTNESS CHECK

As an additional robustness check, a placebo test was conducted using randomly selected non-event dates from the available S&P 500 trading calendar. The purpose of this procedure is to evaluate whether similar CAR patterns could appear around arbitrary dates unrelated to tariff-war events. If placebo windows produce results similar to the observed event windows, the interpretation of the main Event Study should be more cautious. If placebo results are clearly smaller or less structured, the event-based interpretation becomes more credible, although still non-causal.

Table 11: Placebo Robustness Check

Window	Observed Mean CAR	Placebo Mean CAR	Placebo SD	Empirical p-value
$[-1, +1]$	-0.23%	-0.05%	2.02%	0.812
$[-3, +3]$	-0.16%	-0.06%	3.20%	0.910
$[-5, +5]$	0.11%	-0.12%	4.43%	0.956

Notes: Placebo dates were sampled from the real S&P 500 trading calendar using seed 42, excluding real event dates and anchor dates. The empirical placebo p-value is the share of placebo CARs whose absolute value is at least as large as the absolute observed mean CAR.

Table 11 shows that the observed average CAR values are not clearly distinguishable from the placebo distribution. In all three event windows, the empirical placebo p-values are high, ranging from 0.812 to 0.956. This indicates that randomly selected non-event dates frequently generated CAR values at least as large in absolute terms as the observed tariff-war event averages. In practical terms, the placebo test is consistent with the null hypothesis that the observed average CAR values are not statistically exceptional relative to normal market variation. Rather than weakening the thesis, this result clarifies the empirical boundary of the findings: the Event Study documents limited and context-sensitive market reactions, but it does not support a strong claim that tariff-war events produced abnormal returns distinguishable from placebo windows.

4.9 EXPLORATORY MACHINE LEARNING RESULTS

The exploratory Machine Learning layer evaluated whether event-window features contained enough directional signal to classify the sign of market reaction. The majority baseline achieved an accuracy of 55.56%. Logistic Regression achieved an accuracy of 43.06% and an F1 score of 38.81%, while Random Forest achieved an accuracy of 41.67% and an F1 score of 30.00%.

Because the 72 observations are derived from 24 events observed across three event windows, they should not be interpreted as 72 fully independent observations. The same underlying event may contribute multiple rows to the dataset, which reduces the effective independence of the Machine Learning sample. For this reason, the results are interpreted only as an exploratory directional classification diagnostic under small-sample conditions, not as evidence of robust predictive capacity.

Table 12 reports the performance of the exploratory Machine Learning models compared with the majority-class baseline. This comparison is necessary because, in small and imbalanced datasets, a model must be evaluated against a simple benchmark before any predictive interpretation can be considered meaningful.

Table 12: Exploratory Machine Learning Model Performance

Model	Accuracy	F1-score
Majority baseline	55.56%	–
Logistic Regression	43.06%	38.81%
Random Forest	41.67%	30.00%

As shown in Table 12, neither Logistic Regression nor Random Forest outperformed the majority baseline. This result limits the predictive interpretation of the Machine Learning layer and supports its treatment as an exploratory directional classification diagnostic. Rather than weakening the Event Study, this finding suggests that the available event-window features are not sufficient for robust directional classification in the constructed dataset.

The weak predictive performance is also consistent with the difficulty of classifying financial reactions to geopolitical events in small samples. Tariff-war events are heterogeneous, market expectations are difficult to observe directly, and daily returns are influenced by multiple simultaneous factors. For these reasons, the Machine Learning results should be interpreted as evidence of limited predictability in the constructed dataset, not as evidence that tariff-war events were irrelevant.

4.10 SUMMARY OF EMPIRICAL FINDINGS

The empirical record for the United States–China tariff war on the S&P 500 is best described as an initially negative short-window reaction that attenuates or partially reverses in longer windows. The magnitude of those reactions, however, is small: none of the inferential tests reach statistical significance at the 5% level, and the bootstrap confidence intervals include zero in every window. The placebo check adds a further qualification: the observed average CAR values are not clearly distinguishable from those generated around randomly chosen non-event dates, which means the tariff-war timing does not produce statistically exceptional market deviations. The event-level audit confirms that some group averages are dominated by individual observations, with E08 in the $[-3, +3]$ window being the most consequential case.

Escalation and relief categories show different descriptive patterns but no statistically confirmed divergence. The robustness layer shows that the direction of the results is broadly stable across specifications, while the Machine Learning layer shows that the constructed event-window features lack sufficient directional signal for standard clas-

sifiers to improve on a simple majority baseline. The picture that emerges is one of weak, window-sensitive, sample-dependent market associations that are real enough to document but insufficient to support strong causal or predictive claims.



5

5

DISCUSSION

This chapter situates the empirical findings within the literature on trade policy uncertainty, market reaction to information shocks, and event-based financial analysis. Rather than rehearsing the numerical results, the discussion asks what those results reveal about how the U.S. equity market absorbed tariff-war information and where the evidence falls short of stronger claims. The chapter is organized around five main dimensions: the interpretation of market reactions, the distinction between escalation and relief events, the relationship between trade policy uncertainty and volatility, the role of robustness analysis, and the implications of the exploratory Machine Learning layer.

5.1 INTERPRETATION OF MARKET REACTIONS

The results suggest that tariff-war events operated as relevant information events for the U.S. stock market. This interpretation is consistent with the logic of the Event Study framework, according to which asset prices may react when new information affects expectations about future cash flows, uncertainty, or risk. In the context of the United States–China tariff war, announcements, implementations, suspensions, and negotiation signals may have changed investors' expectations about trade costs, supply chain disruptions, corporate margins, and geopolitical risk.

The negative short-window reaction observed in the results is consistent with an initial risk-averse response to trade-related uncertainty. Tariff-war news can increase ambiguity about future economic conditions, particularly when investors are unsure about the scope, duration, or intensity of policy escalation. However, the attenuation or partial reversal observed in longer windows suggests that market reactions were not necessarily persistent in the same direction. This may indicate that investors reassessed the information after the initial shock, incorporated additional expectations, or corrected an initial overreaction.

Financial markets process public information quickly, but not always in a simple or linear way. The Event Study literature supports the use of abnormal returns to evaluate market responses to information arrivals, while also requiring caution when interpreting those responses as causal effects [9–11]. The observed CAR patterns in this thesis are therefore best read as evidence of temporal association between tariff-war events and market behavior, not as proof that each event individually drove the observed price

movement.

The absence of statistically significant cumulative abnormal returns across all three event windows, corroborated by the placebo robustness check, constitutes an empirically meaningful finding in itself. Rather than indicating a methodological failure, this result is consistent with the semi-strong form of market efficiency [27], which posits that publicly available information (including trade policy announcements) is rapidly incorporated into asset prices prior to or at the moment of the event. Under this interpretation, the limited evidence of abnormal market behavior suggests that the U.S. equity market had already priced in much of the tariff-war information by the time events became formally observable in the daily trading calendar. This interpretation is reinforced by the nature of the conflict: the U.S.–China tariff war unfolded gradually over multiple years, providing investors with sustained exposure to escalation signals, negotiation rhetoric, and policy reversals. As a result, individual events may have carried limited new informational content relative to what market participants had already embedded in their expectations. The contribution of this study therefore lies not in demonstrating a strong causal market response, but in providing a transparent, reproducible empirical framework that rigorously documents the absence of systematic abnormal returns, a finding that is equally important for understanding how geopolitical trade shocks are absorbed by mature equity markets.

The comparison with related empirical studies also supports a cautious interpretation of the findings. Studies using firm-level, sector-level, value-chain, volatility, or risk-channel approaches often identify stronger or more specific effects of the United States–China tariff war [4, 25, 26]. This does not contradict the results of this thesis. Instead, it clarifies the scope of the evidence. The present research evaluates whether tariff-war events generated a strong and systematic reaction in the aggregate daily S&P 500. The absence of statistically strong average CAR patterns in this setting should therefore be interpreted as evidence about the limits of aggregate index-level detection, not as evidence that tariff-war events were financially irrelevant.

5.2 ESCALATION AND RELIEF DYNAMICS

Comparing escalation and relief events adds texture to the aggregate picture. Escalation events encompass actions that intensify trade tensions, such as tariff announcements, tariff increases, implementations, and retaliatory measures. Relief events encompass suspensions, reductions, agreements, or negotiation-related developments that reduce or temporarily ease trade tensions. The results suggest that these categories did not produce perfectly symmetric market responses.

The positive average CAR values observed for escalation events in the wider windows warrant careful interpretation. They do not imply that escalation benefited the market. A more cautious reading is that some escalation events may have resolved

uncertainty about the policy path, confirmed information investors had already priced in, or coincided with offsetting market forces that diluted the negative content of the tariff signal. Because the available data do not allow the thesis to directly observe investor expectations before each event, this explanation should be treated as a plausible interpretation rather than as direct evidence.

The relief group calls for analogous caution. Its average CAR values are pulled down by adverse observations, most notably E08 in the $[-3, +3]$ window, which unfolded during a broader period of market stress in December 2018 and produced one of the most negative values in the event-level audit table. The negative average CAR for relief events should therefore not be read as evidence that relief measures failed to reassure markets. The positive median CAR values within the relief group indicate that the majority of relief events did not generate negative reactions; the mean is distorted by a small number of extreme observations.

Escalation and relief, then, are best understood as analytical categories used to organize short-window market reactions, not as direct labels of positive or negative economic value. The group-level comparison remains descriptive and useful, but it is sensitive to individual events and should not be taken as evidence of statistically definitive asymmetry.

5.3 TRADE POLICY UNCERTAINTY AND FINANCIAL INSTABILITY

The present results align with literature identifying trade policy uncertainty as a relevant source of market instability. Baker, Bloom, and Davis show that policy uncertainty can affect economic expectations and investment behavior, while Caldara et al. demonstrate that trade policy uncertainty can produce measurable economic effects [2, 3]. Here, tariff-war events are associated with abnormal returns and periods of localized volatility, especially during the core phase of the conflict.

This relationship should nonetheless be interpreted carefully. The analysis does not establish that tariff-war events were the only or dominant cause of elevated volatility in the observed periods: monetary policy, macroeconomic releases, corporate earnings, and other geopolitical developments operate simultaneously. The study therefore reads the results as evidence that tariff-war events coincided with measurable market reactions, not as proof that trade policy uncertainty alone drove market behavior.

The volatility analysis adds a risk dimension that CAR alone cannot capture. When abnormal returns occur alongside higher local volatility, it suggests that tariff-related information may have shifted both price direction and investors' risk perception. Trade policy shocks, in other words, may affect not only expected returns but also the degree of uncertainty investors assign to those expectations, a distinction that matters for interpreting why market reactions around certain events appear larger than others.

5.4 ROBUSTNESS INTERPRETATION

The robustness and sensitivity analyses help evaluate whether the main findings depend excessively on a single methodological specification. By varying estimation windows, signal thresholds, and sample composition, the study examines whether the interpretation remains reasonably stable across plausible analytical choices. This is especially important in small event samples, where results may be sensitive to individual events or parameter choices.

The robustness results suggest that the main interpretation is not entirely driven by one specific configuration. Although the magnitude of CAR estimates changes across specifications, the broader pattern of short-window reaction, window sensitivity, and sample-dependent attenuation remains visible. However, the placebo robustness check shows that the observed average CAR values are not statistically exceptional relative to randomly selected non-event windows. Robustness should therefore be read as evidence of internal stability across selected specifications, not as evidence of strong or causal market effects.

Robustness checks and placebo checks answer different questions, and keeping them separate matters. The robustness analysis asks whether the results change sharply when reasonable methodological parameters are modified; the placebo test asks whether the observed CAR patterns are unusual relative to random non-event dates. In this thesis, the first answer is moderately reassuring, while the second is deliberately limiting: the results are organized and internally consistent, but they are not statistically exceptional when benchmarked against the normal variation of daily returns.

This approach is consistent with the broader principles of empirical financial research, where model specification, sample construction, and sensitivity testing are central to the credibility of conclusions [28, 29]. In this thesis, robustness functions as a stability check rather than as a definitive validation mechanism.

The robust central-tendency estimators reinforce this interpretation. Because the median, the trimmed mean, and the winsorized mean turn mildly positive from the $[-3, +3]$ window onward, whereas the mean remains slightly negative in the $[-3, +3]$ window, the aggregate negative mean in that window is best understood as the footprint of a few high-impact events rather than as evidence of a systematic negative reaction. The leave-one-out analysis makes this concrete: removing a single event, E08, is enough to change the sign of the $[-3, +3]$ mean. This does not strengthen the case for a causal market effect; on the contrary, it reinforces the central message that the observed CAR patterns are weak, heterogeneous, sensitive to individual observations, and statistically indistinguishable from zero.

5.5 MACHINE LEARNING INTERPRETATION

The exploratory Machine Learning results provide a useful but limited complement

to the Event Study. Logistic Regression and Random Forest did not outperform the majority baseline, which indicates that the constructed event-window features were not sufficient to classify the direction of market reaction with reliability. This result does not weaken the Event Study findings. Instead, it clarifies that the observed abnormal-return patterns are difficult to translate into predictive rules in a small and noisy financial dataset.

Several factors explain why the models struggled. The dataset contains 72 observations derived from 24 events across three windows (not 72 independent events), which substantially limits effective sample size. Tariff-war events are also intrinsically heterogeneous: announcements, implementations, suspensions, and agreements may carry very different informational content depending on timing and context. Market reactions, furthermore, depend on prior expectations that are not directly observable; an event that appears negative in policy terms may generate little reaction if investors had already priced it in.

On these grounds, the Machine Learning component should be understood as an exploratory directional classification diagnostic, not as an inferential or predictive contribution on its own. Its purpose is to evaluate whether the features constructed from the Event Study pipeline contain enough directional signal for standard classifiers to outperform a simple majority baseline. Since the models did not do so, the result confirms that the weak temporal associations observed in the CAR analysis do not translate into robust learnable patterns under the available small-sample conditions. This supports the decision to preserve the Event Study as the central empirical method and to interpret the Machine Learning layer only as a supplementary diagnostic of predictability limits.

5.6 METHODOLOGICAL CONTRIBUTION

The main methodological contribution of this thesis is the construction of an empirical framework that is easier to inspect and replicate for analyzing financial reactions to trade-policy shocks. The contribution does not lie in applying the most complex model possible. Rather, it lies in integrating a classical Event Study framework with structured data processing, robustness analysis, volatility assessment, exploratory Machine Learning, and validation procedures [13, 14].

This architecture is valuable because it allows the research to move from historical events to financial inference in a documented and transparent process. The event chronology defines the empirical object. Market data and financial features create the quantitative foundation. Event windows, expected returns, abnormal returns, and CAR form the analytical core. Robustness and volatility layers support interpretation. The Machine Learning layer tests whether the constructed features carry directional classification information. Validation procedures help ensure that the pipeline is designed to

make each analytical step visible and transparent.

This design also responds to a common challenge in applied empirical research: avoiding disconnected methodological blocks. In this thesis, each layer has a specific function within a single framework. The Event Study measures market reaction; robustness evaluates stability; volatility adds a risk dimension; Machine Learning tests predictive capacity; and validation supports replicability. The result is a cohesive structure that can be inspected, extended, or adapted in future research.

5.7 PRACTICAL AND ACADEMIC IMPLICATIONS

For researchers, the study demonstrates the usefulness of combining established empirical finance methods with replicable data science workflows. Event studies remain appropriate for measuring market reactions to dated information shocks, but their credibility improves when supported by transparent data construction, sensitivity analysis, and validation routines. The thesis also shows that Machine Learning should be integrated carefully in small financial datasets, especially when the goal is interpretation rather than prediction.

For investors and policy analysts, the findings suggest that trade-policy events should be interpreted within a broader context of expectations and uncertainty. Market reactions may depend not only on whether a policy action appears aggressive or conciliatory, but also on whether the event was anticipated, whether it resolves uncertainty, and whether it changes expectations about future policy direction. This helps explain why escalation and relief events may not generate symmetric responses.

Academically, the thesis contributes to the discussion on how geopolitical and trade-policy shocks can be studied through a structured empirical framework. It reinforces the importance of cautious interpretation: tariff-war events are associated with measurable market reactions, but these reactions are probabilistic, context-dependent, and influenced by broader market conditions. By combining Event Study methodology, robustness analysis, and methodologically transparent computational practices, the research offers a framework that can be expanded in future studies involving other markets, sectors, or policy conflicts.



6

6

CONCLUSION

6.1 RESEARCH OBJECTIVE REVISITED

This thesis investigated market reactions to key events of the United States–China tariff war in the U.S. stock market, with primary emphasis on the S&P 500. Using a structured Event Study methodology complemented by volatility analysis, robustness checks, and an exploratory Machine Learning layer, the general objective was to evaluate whether and how the S&P 500 reacted to key tariff-war events.

Each objective established in the introduction was addressed as follows.

General objective. Cumulative abnormal returns were estimated for three event windows across three event groups, and the results were interpreted in light of robustness tests, placebo validation, and robust central-tendency estimators. The evidence indicates limited, window-sensitive, and heterogeneous market associations: real enough to document, insufficient to support strong causal claims.

Specific objective 1 (historical chronology). The master chronology comprises 67 documented events organized into escalation, relief, and structural categories, spanning core, pandemic, and post-pandemic regimes from 2018 to 2026.

Specific objective 2 (abnormal returns and CAR). Formal CAR values were estimated for all 24 market-covered events across the $[-1, +1]$, $[-3, +3]$, and $[-5, +5]$ windows, with full inferential results reported in Chapter 4.

Specific objective 3 (regime separation). The three-phase classification and the market coverage filter delimited the core phase (17 events) and the expanded market-covered sample (24 events) for financial inference, while the post-pandemic phase (31 events) was documented historically.

Specific objective 4 (robustness). Three robustness dimensions were evaluated: alternative estimation windows, alternative signal thresholds, and sample composition comparisons. A sign test, leave-one-out analysis, and placebo test further assessed the stability of the main interpretation.

Specific objective 5 (Machine Learning). Logistic Regression and Random Forest classification models were implemented. Neither outperformed the majority baseline, an informative result in itself: the constructed event-window features did not contain sufficient directional signal for reliable classification under small-sample conditions.

Rather than claiming strict econometric causality, the thesis examined temporal associations between historically dated tariff-war events and financial market behavior. That framing reflects a genuine methodological boundary: market prices are shaped by many simultaneous economic, political, and financial forces that cannot be fully separated within this design.

6.2 MAIN FINDINGS

The empirical results point to limited, context-sensitive market reactions. S&P 500 cumulative abnormal returns were negative in the shorter windows and slightly positive in the widest, tracing an initial adverse reaction that attenuated or partially reversed as the event horizon expanded. This pattern indicates that market responses to tariff-war events were sensitive to the chosen event horizon. Short windows captured the immediate reaction to tariff-related information, while longer windows suggested a process of market absorption, correction, or reassessment.

The comparison between escalation and relief events showed that these groups displayed different patterns of cumulative abnormal returns. This does not imply that escalation events were beneficial or that relief events were ineffective. Instead, the results suggest that market reactions depended on the type of tariff-related signal, the degree of investor anticipation, and the uncertainty surrounding each event. Escalation and relief should therefore be interpreted as analytical categories rather than direct measures of economic desirability.

The robustness and sensitivity analyses supported the stability of the central interpretation, although the magnitude of the results varied across specifications. This indicates that the findings were not entirely driven by a single estimation window, signal threshold, or sample composition. At the same time, these tests do not eliminate the need for cautious interpretation, especially given the limited size of the event sample and the presence of possible confounding factors.

A robust-estimation layer (comprising the median, the trimmed and winsorized means, the sign test, and a leave-one-out influence analysis) added important texture to these conclusions. The robust estimators turned mildly positive in the wider windows while remaining statistically indistinguishable from zero, and the leave-one-out analysis showed that the slightly negative mean in the $[-3, +3]$ window depended strongly on a single high-impact event. The aggregate market reactions documented in this thesis are therefore weak, window-sensitive, and vulnerable to individual observations: patterns worth documenting, but not a basis for strong generalization.

The Machine Learning results reinforce this picture. Logistic Regression achieved an accuracy of 43.06% and an F1 score of 38.81%; Random Forest achieved 41.67% and 30.00%, respectively. Both fell below the majority baseline of 55.56%, confirming that the event-window features did not contain sufficient directional signal for reliable

classification. Machine Learning in this thesis is therefore diagnostic, not evidentiary.

6.3 CONTRIBUTIONS

This research contributes to empirical finance by applying a structured Event Study framework to the United States–China tariff war and by organizing the analysis through a transparent data science pipeline. The study integrates event chronology construction, market data processing, financial feature engineering, event window design, abnormal return estimation, CAR aggregation, volatility analysis, robustness auditing, and exploratory Machine Learning into a single methodological architecture.

A central contribution of the thesis is the separation between the master chronology, the core sample, and the expanded market-covered sample. The master chronology documents 67 tariff-war events, while the core Event Study sample focuses on 17 primary events and the expanded market-covered analysis includes 24 events with sufficient financial data. This structure improves the historical completeness of the project without overstating the scope of financial inference.

The study also contributes methodologically by emphasizing reproducibility. The use of modular datasets, validation procedures, structured outputs, and replicable scripts increases the transparency of the empirical process. This approach is consistent with the principles of verifiable computational research and helps make the analysis reviewable, extendable, and less vulnerable to hidden processing errors [13, 14].

Finally, the exploratory Machine Learning layer contributes by showing the limits of direct classification in this specific setting. The fact that Logistic Regression and Random Forest did not outperform the majority baseline is informative because it suggests that tariff-war market reactions are difficult to classify using a small event-window dataset. This reinforces the importance of maintaining the Event Study as the central empirical method.

6.4 LIMITATIONS

Several limitations should be acknowledged. First, although the Event Study framework successfully isolates abnormal returns around specific dates, it does not establish strict econometric causality. Other economic and financial developments occurring in close temporal proximity to the tariff-war dates may have shaped market behavior, and their potential confounding influence cannot be fully disentangled within the present design.

Second, the study relies on daily market data. While appropriate for the scope of this thesis, daily data do not capture intraday reactions or the precise timing of market adjustment within a single trading session. Future research employing intraday data could offer a more granular view of price discovery around tariff-war announcements, illuminating how news is transmitted into asset prices within minutes or hours of a policy

event.

Third, the event sample is limited. Although the master chronology documents 67 events, financial inference is restricted to those with sufficient market coverage. This constraint is especially relevant for the post-pandemic regime, where events remain documented but are not fully incorporated into the event study inference due to the available market dataset. The small sample also renders the results sensitive to individual observations. For instance, the event-level audit table shows that E08 produced a large negative CAR in the $[-3, +3]$ window, and the leave-one-out analysis quantifies how strongly this single observation affects the aggregate mean. This pattern reinforces the need to interpret group averages and window-level results with caution, since individual events can meaningfully influence aggregate patterns in a relatively small Event Study sample.

Fourth, the expected return model used in the thesis is intentionally simple. The mean-adjusted return model offers transparency and interpretability, but alternative specifications (such as the market model, the Capital Asset Pricing Model, or Fama–French multifactor models) could provide richer risk adjustment in future research and may surface effects that the current model leaves unaccounted for.

Finally, the Machine Learning layer is exploratory in nature and constrained by sample size. The models deployed in the thesis did not outperform the majority baseline, which limits their predictive interpretation. Accordingly, the ML results should be read as evidence about the difficulty of classifying market reactions within this dataset rather than as an indictment of the event study framework itself.

6.5 FUTURE RESEARCH

Future research could expand this framework in several directions. First, the analysis could incorporate a broader set of international indices to evaluate whether the market reaction to the United States–China tariff war differed across regions. Including European, Asian, and emerging market indices would allow researchers to examine whether tariff-war shocks produced asymmetric effects across global equity markets.

Second, future studies could conduct sector-level analysis. The S&P 500 provides a broad view of the U.S. stock market, but tariff exposure is not distributed equally across sectors. Industries such as technology, manufacturing, agriculture, consumer goods, and logistics may react differently to trade-policy shocks. A sector-level event study could therefore provide a more detailed understanding of how trade policy uncertainty propagates through the economy.

Third, future research could compare alternative expected return models. The present thesis uses a mean-adjusted return model for transparency and methodological simplicity. Because the object of analysis is the S&P 500 itself, the expected return is an internal pre-event baseline derived from the same index; no external benchmark

absorbs broad market-wide movements that may coincide with event dates. Future work could test whether the estimated abnormal returns change when a market model uses an external reference series (such as a global equity index excluding the United States, an MSCI World ex-US proxy, or a developed-market benchmark) as the control series. Testing the market model, CAPM, or Fama-French factor specifications in this context would reveal whether the observed CARs are sensitive to the choice of baseline and whether outlier events such as E08 retain their magnitude once a broader macro-financial context is controlled for [30].

Fourth, future research could extend the market dataset through 2026. This would allow a more complete analysis of the pandemic and post-pandemic regimes and would make it possible to evaluate whether the dynamics observed during the classical tariff-war phase persisted in later periods.

Fifth, future studies could incorporate Natural Language Processing to measure the rhetorical dimension of trade policy announcements. The present thesis does not perform textual analysis, but future work could use sentiment analysis or policy text classification to distinguish between rhetorical threats, negotiation signals, and formal tariff implementations [31].

Finally, future research could explore regime-switching models to identify endogenous market states. In this thesis, regimes are defined historically according to the structure of the tariff-war chronology and the pandemic period. Future work could compare this historical regime separation with statistically inferred regimes, such as Markov-switching models, to evaluate whether market volatility states align with the historical phases of the conflict [32, 33].

6.6 FINAL STATEMENT

This thesis examined whether key events of the United States–China tariff war were associated with measurable reactions in the U.S. stock market. The evidence leads to an honest, if constrained, conclusion: tariff-war events left a detectable short-window footprint in S&P 500 returns, but that footprint was too small, too sensitive to individual events, and too dependent on window choice to constitute statistically strong evidence of a systematic effect. The robust-estimation layer reinforced this reading by showing that the mean in the $[-3, +3]$ window, the specification most often cited in tariff-war studies, reverses sign when a single influential event is removed. This is not a failure of the thesis; it is a precise empirical finding: at the level of daily aggregate equity returns, individual tariff-war events did not produce a market reaction strong enough to be reliably distinguished from normal variation.

The framework developed in this thesis combines Event Study methodology, structured data processing, robustness analysis, volatility assessment, and exploratory Machine Learning. Its value lies in documenting both what the evidence suggests and

what it cannot support, an equally informative pair of statements in a literature where the pressure to report positive results is well known. Methodological transparency is not a hedge; it is the condition under which the reported findings can be taken at face value.

The United States–China tariff war reshaped trade relationships, supply chain expectations, and geopolitical risk perceptions. This thesis adds to that discussion by showing that, when examined through daily equity-market data, the S&P 500's reaction to individual tariff-war events was limited and highly dependent on context. Future research may expand this framework to other markets, sectors, policy conflicts, and higher-frequency data, but the central contribution remains the same: careful empirical work should report uncertainty, outliers, and limitations as clearly as it reports results.

The background image shows a modern building interior with curved balconies and a central courtyard. The balconies have glass railings and are illuminated with warm lights. The courtyard features several large, cylindrical planters with green plants. The overall atmosphere is clean and contemporary.

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APPENDICES

Appendix A - Event Audit Table

This appendix reports the event-level audit table used to support the empirical analysis. Its purpose is to make the construction of the event sample transparent and auditable. The tables distinguish between the broader historical chronology of tariff-war events and the subset of events that could be used for financial inference. For each event, the appendix reports the original event date, the market anchor date when available, a short event description, the event group, the historical regime, the inclusion status, and the reason for exclusion when applicable. This distinction is important because the thesis documents 67 events, but only events with sufficient market coverage and valid event-window construction are used in the Event Study calculations.

The appendix is divided into two parts. Tables 13–16 report the full event chronology and inclusion status. Tables 17–18 report the formal S&P 500 CAR values for the 24 events included in financial inference. Together, these tables allow the reader to verify which events were documented, which events entered the empirical analysis, and how the event-level CAR values were distributed across the three event windows.

Table 13: Event Chronology and Inclusion Status (Part 1)

Event ID	Event date	Anchor date	Short description	Group	Regime	Included / reason
E01	2018-03-22	2018-03-22	US announces Section 301 tariff action on Chinese imports	escalation	core	yes / n.a.
E02	2018-04-06	2018-04-06	USTR publishes proposed tariff list targeting Chinese imports	escalation	core	yes / n.a.
E03	2018-06-15	2018-06-15	USTR finalizes 25 percent tariffs on 50 billion dollars of Chinese goods	escalation	core	yes / n.a.
E04	2018-07-06	2018-07-06	First tranche of bilateral tariffs enters into force	escalation	core	yes / n.a.
E05	2018-07-11	2018-07-11	USTR proposes tariffs on an additional 200 billion dollars of Chinese goods	escalation	core	yes / n.a.
E06	2018-08-23	2018-08-23	Second tranche of bilateral tariffs enters into force	escalation	core	yes / n.a.
E07	2018-09-24	2018-09-24	US implements tariffs on 200 billion dollars of Chinese imports	escalation	core	yes / n.a.
E08	2018-12-19	2018-12-19	US postpones scheduled increase of tariffs	relief	core	yes / n.a.
E09	2019-03-05	2019-03-05	US extends suspension of tariff increase	relief	core	yes / n.a.
E10	2019-05-10	2019-05-10	US raises tariffs from 10 percent to 25 percent	escalation	core	yes / n.a.
E11	2019-08-20	2019-08-20	USTR formalizes new tariffs on Chinese imports	escalation	core	yes / n.a.
E12	2019-08-30	2019-08-30	US raises planned tariff rate	escalation	core	yes / n.a.
E13	2019-09-01	2019-09-03	New tariffs enter into force	escalation	core	yes / n.a.
E14	2019-12-13	2019-12-13	Phase One agreement announced	relief	core	yes / n.a.
E15	2019-12-18	2019-12-18	US suspends planned tariffs	relief	core	yes / n.a.
E16	2020-01-15	2020-01-15	Phase One agreement signed	relief	core	yes / n.a.
E17	2020-02-14	2020-02-14	US reduces tariffs	relief	core	yes / n.a.
E18	2020-02-17	2020-02-18	China introduces tariff exemption process	relief	pandemic	yes / n.a.
E19	2020-07-14	2020-07-14	China increases US agricultural imports	relief	pandemic	yes / n.a.
E20	2020-07-22	2020-07-22	US considers tariff exclusions	relief	pandemic	yes / n.a.

Notes: Source-level metadata were not available in the project dataset and are therefore not displayed as a separate column. The term “n.a.” means not applicable.

Table 14: Event Chronology and Inclusion Status (Part 2)

Event ID	Event date	Anchor date	Short description	Group	Regime	Included / reason
E21	2020-08-11	2020-08-11	US changes Hong Kong trade labeling rules	escalation	pandemic	yes / n.a.
E22	2020-08-15	2020-08-17	Trade review postponed	structural	pandemic	yes / n.a.
E23	2020-08-19	2020-08-19	US suspends tax exemptions	escalation	pandemic	yes / n.a.
E24	2020-08-25	2020-08-25	Trade talks continue	relief	pandemic	yes / n.a.
E25	2021-05-01	n.a.	China reduces tariffs on steel	structural	pandemic	no / outside validated market-data scope
E26	2021-07-01	n.a.	China reduces tariffs on IT goods	structural	pandemic	no / outside validated market-data scope
E27	2021-10-08	n.a.	US considers tariff exclusions	relief	pandemic	no / outside validated market-data scope
E28	2021-10-12	n.a.	US reinstates tariff exclusions	relief	pandemic	no / outside validated market-data scope
E29	2022-01-01	n.a.	China adjusts tariffs	structural	pandemic	no / outside validated market-data scope
E30	2022-03-23	n.a.	US reinstates exclusions	relief	pandemic	no / outside validated market-data scope
E31	2022-05-01	n.a.	China reduces tariffs on coal	structural	pandemic	no / outside validated market-data scope
E32	2022-05-05	n.a.	US initiates tariff review	escalation	pandemic	no / outside validated market-data scope
E33	2022-12-21	n.a.	US extends exclusions	relief	pandemic	no / outside validated market-data scope
E34	2023-01-01	n.a.	China reduces tariffs on consumer goods	structural	pandemic	no / outside validated market-data scope
E35	2023-02-07	n.a.	US tariffs on washing machines expire	relief	pandemic	no / outside validated market-data scope
E36	2023-09-22	n.a.	US China economic groups formed	structural	pandemic	no / outside validated market-data scope
E37	2024-04-17	n.a.	US proposes higher tariffs on metals	escalation	post	no / outside validated market-data scope
E38	2024-05-14	n.a.	US proposes tariffs on EVs and semiconductors	escalation	post	no / outside validated market-data scope
E39	2024-05-30	n.a.	US extends tariff exclusions	relief	post	no / outside validated market-data scope
E40	2024-09-13	n.a.	US finalizes tariff increases	escalation	post	no / outside validated market-data scope

Notes: Events marked as outside validated market-data scope remain documented in the master chronology but are not used in the financial Event Study calculations.

Table 15: Event Chronology and Inclusion Status (Part 3)

Event ID	Event date	Anchor date	Short description	Group	Regime	Included / reason
E41	2024-09-14	n.a.	China signals retaliation	retaliation	post	no / outside validated market-data scope
E42	2024-09-24	n.a.	US proposes further tariff adjustments	escalation	post	no / outside validated market-data scope
E43	2024-09-27	n.a.	New tariffs implemented	escalation	post	no / outside validated market-data scope
E44	2024-11-22	n.a.	China protests US actions	escalation	post	no / outside validated market-data scope
E45	2025-01-01	n.a.	US increases semiconductor tariffs	escalation	post	no / outside validated market-data scope
E46	2025-01-20	n.a.	New administration shifts trade stance	structural	post	no / outside validated market-data scope
E47	2025-02-04	n.a.	US imposes broad tariffs	escalation	post	no / outside validated market-data scope
E48	2025-02-10	n.a.	China retaliates	retaliation	post	no / outside validated market-data scope
E49	2025-03-04	n.a.	US increases tariffs again	escalation	post	no / outside validated market-data scope
E50	2025-03-10	n.a.	China responds again	retaliation	post	no / outside validated market-data scope
E51	2025-04-10	n.a.	Tariffs reach peak levels	escalation	post	no / outside validated market-data scope
E52	2025-05-14	n.a.	Tariff reduction agreement	relief	post	no / outside validated market-data scope
E53	2025-05-31	n.a.	US extends exclusions	relief	post	no / outside validated market-data scope
E54	2025-08-12	n.a.	Temporary trade truce	relief	post	no / outside validated market-data scope
E55	2025-09-01	n.a.	China imposes duties	retaliation	post	no / outside validated market-data scope
E56	2025-10-30	n.a.	Leaders agree to adjust measures	relief	post	no / outside validated market-data scope
E57	2025-11-05	n.a.	China removes tariffs	relief	post	no / outside validated market-data scope
E58	2025-11-07	n.a.	China suspends export controls	relief	post	no / outside validated market-data scope
E59	2025-11-09	n.a.	China suspends restrictions	relief	post	no / outside validated market-data scope
E60	2025-11-10	n.a.	Mutual tariff reduction agreement	relief	post	no / outside validated market-data scope

Notes: These events are retained for historical completeness and regime documentation, but they are excluded from financial inference due to lack of market coverage in the dataset.

Table 16: Event Chronology and Inclusion Status (Part 4)

Event ID	Event date	Anchor date	Short description	Group	Regime	Included / reason
E61	2026-01-01	n.a.	US increases tariffs on strategic goods	escalation	post	no / outside validated market-data scope
E62	2026-02-24	n.a.	New tariff framework begins	structural	post	no / outside validated market-data scope
E63	2026-03-01	n.a.	China updates trade law	structural	post	no / outside validated market-data scope
E64	2026-03-03	n.a.	Research highlights trade decoupling	structural	post	no / outside validated market-data scope
E65	2026-03-16	n.a.	China urges tariff removal	retaliation	post	no / outside validated market-data scope
E66	2026-03-27	n.a.	China launches investigations	retaliation	post	no / outside validated market-data scope
E67	2026-04-08	n.a.	Data on export controls released	structural	post	no / outside validated market-data scope

Notes: The final events in the chronology are documented for transparency but are not included in the Event Study because they fall outside the available market-covered period.

The following tables report the event-level formal CAR values for the events included in financial inference. These values show the cumulative abnormal return of the S&P 500 around each event for the three event windows used in the thesis. They provide a direct audit trail between the event chronology and the aggregate CAR results discussed in the Results chapter.

Table 17: Event-Level Formal CAR Values (Part 1)

Event ID	CAR $[-1, +1]$	CAR $[-3, +3]$	CAR $[-5, +5]$
E01	-5.06%	-5.69%	-4.86%
E02	-0.76%	3.33%	3.56%
E03	-0.67%	-1.94%	-2.77%
E04	2.70%	2.29%	4.10%
E05	0.48%	2.18%	3.12%
E06	0.22%	1.20%	2.20%
E07	-0.74%	-0.17%	-0.11%
E08	-2.75%	-11.02%	-4.21%
E09	-1.87%	-3.44%	-2.77%
E10	-2.82%	-3.87%	-3.68%
E11	1.61%	1.15%	1.00%
E12	0.91%	3.99%	2.94%
E13	0.97%	4.96%	6.44%
E14	1.30%	1.10%	2.27%
E15	0.22%	1.24%	2.60%
E16	0.51%	0.55%	1.36%
E17	-0.70%	-0.37%	-5.15%
E18	0.10%	-1.20%	-7.03%
E19	0.74%	0.41%	0.37%
E20	-0.40%	0.96%	2.24%

Notes: CAR refers to the formal cumulative abnormal return of the S&P 500. Values are shown only for events included in financial inference.

Table 18: Event-Level Formal CAR Values (Part 2)

Event ID	CAR $[-1, +1]$	CAR $[-3, +3]$	CAR $[-5, +5]$
E21	-0.22%	-1.21%	-1.17%
E22	-0.35%	-0.38%	-0.66%
E23	-0.60%	0.06%	1.69%
E24	1.54%	1.92%	1.12%

Notes: CAR refers to the formal cumulative abnormal return of the S&P 500. Values are shown only for events included in financial inference.



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